Algorithm Benchmarking and Outcomes

The Society Connectedness & Health Lab
Yale School of Public Health

Church Closures Dashboard Project

SOCAH
SOCIETY CONNECTEDNESS AND HEALTH LAB

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Preparation of the 2026 Format Raw Data

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Overview

The project objectives were as follows:

1. **REVIEW:** Evaluate the raw data and previously outlined cleaning, validation, and harmonization methods employed by Dr. Insang Song for accuracy and identify opportunities for improvement.
2. **PREPARATION:** Implement identified improvements to prepare the dataset for analysis and distribution to qualifying collaborators.
3. **METRICS:** Calculate the metrics necessary for dynamic visualization.

Additionally, the project Principal Investigator (PI), Professor Yusuf Ransome, requested the pipeline be reproducible for future collaborators and readily adaptable to potential changes in project scope.

Following receipt of the raw data in Summer 2025, a review was conducted and findings were compared against the original methodology provided by Dr. Song. When an updated dataset was subsequently delivered in May 2026, the pipeline was expanded to accommodate two designated variations: the **2023 Format** and the **2026 Format**.

This document provides an overview of the methodological changes implemented following the initial prototype developed using the 2023 Format, and to accommodate the differences identified in the 2026 Format Review. While many of the fundamental methods were retained, notable changes to the step-wise ordering and consideration of edge cases were implemented in light of the prototype's algorithm performance and access to the High Performance Computer (HPC).

Results generated from this pipeline were used for the Fall 2026 dashboard developed by Gordon Tu.

Important Considerations

Selected code excerpts are referenced below. All snippets are drawn from scripts saved to the "Code/" directory, each prefixed by a descriptor indicating the step to which they apply:

Clean Raw Data_Step *: A discrete step in the methods pipeline outlined below. Additional classifiers may indicate whether the script was designed for use on the HPC and/or whether it applies to the 2023 or 2026 version of the raw data.

Ignored Files

Individual-level files are withheld from GitHub and Git tracking in accordance with the Data Use Agreement (DUA) with the data provider. Many of these are stored in `**/KEEP LOCAL/` directories or are explicitly listed under the "Project-Specific Exclusions" section of the `.gitignore` file.

Most files are available on the SOCAH Lab OneDrive for the project, while others are user-specific and do not need to be shared. If there are local-only files for this project, they will be added to the "Church-Closures-Dashboard GitHub LOCAL ONLY Files" directory. This directory will contain `README.md` files specifying the intended use and filepath for each file.

If you do not have access to the SOCAH Lab OneDrive, carefully follow the codebase instructions to establish the repository and replicate results on your own device. Raw data access instructions may vary depending on your institution.

Environment Reproducibility

Results were produced using R (v4.5.2), RStudio (v2026.01.1+403), and Quarto (v1.8.27). The `renv` package (v1.1.4) is included to reproduce the coding environment, capturing all relevant packages and their versions. If issues arise when running the scripts, verify that the environment is initialized and that the correct versions of R, RStudio, and Quarto are being used.

> **Compatibility**
>
> Updating packages, R, or Quarto may introduce compatibility issues with legacy code that will need to be resolved on a case-by-case basis.

First-time users should initialize the environment to install all required packages and ensure a reproducible coding environment. To do so:

1. In the command-line application (i.e., Terminal for Macs or Git Bash for Windows) navigate to the codebase file location.

Command-Line Interface

```
cd "FILE-PATH/Church-Closures-Dashboard/"
```

2. Launch the project by opening the `*.Rproj` file in RStudio.
3. In the R console, activate the environment by running the following lines of code:

Command-Line Interface

```
renv::restore()   # Download packages and their version saved in the
↪  lockfile.
```

> **Note**
>
> If you are asked to update packages, say no. `renv` is intended to recreate the same environment under which the project was created, making it reproducible. You are ready to proceed when running `renv::restore()` gives the output:

RStudio Console

```
- The library is already synchronized with the lockfile.
```

If you experience any trouble with this step, you might want to confirm that you are using R (v4.5.2) in the RStudio (v2026.01.1+403) and Quarto (v1.8.27). You can also read more about `renv` in their vignette [1].

Methods

Approach and Methodological Objectives

The following pipeline was designed to address two primary goals:

1. Produce a cleaned and validated form of the raw data suitable for dynamic visualization at the finest regional resolution: block groups.
2. Associate each address with census boundaries defined during the 2000, 2010, and 2020 decennial years, including block group, tract, county, state, and ZIP Code Tabulation Areas (ZCTAs).

Documentation on the expected accuracy of geocoordinates provided in the raw data was not made available by the data distributor. It was therefore considered important to compare the raw geolocations against a trusted database in order to associate verified geolocations and assess the extent of error in the raw form. While not the most accurate approach, address-based database matching is a generally accepted method for confirming approximate geocoordinates [2].

Each address's geolocation coordiantes can be used to annotate each entry with the respective decennial years census boundary via point-in-polygon geocoding. The U.S. Census Bureau TIGER/Line shapefiles containing the required census boundary information provide corresponding files for decennial years requested: 2000, 2010, and 2020. However, this technique can be computationally costly; resulting in maxing in memory limits or causing preclusively slow loading if not handled strategically.

During the raw data review, it was noted that typographical errors resulted in duplicate address entries, where the associated open/closure years remained unique. Ideally, these errors

would be cleanly reconciled to produce a concise dataset free of unnecessary redundancies. However, the presence of address variants is not precluding to the overall goals of this project. Mitigating these variations is nonetheless worthwhile to minimize unnecessary computational costs during geocoding and subsequent metric summarization across all addresses. Therefore, an attempt will be made to responsibly consolidate addresses using string similarity matching. Additional resolution is expected when address validation is done against a trusted database.

Limitations on Pre-Cleaning and Data Retention

It is important that simplifying the data cleaning and validation process does not necessitate the removal of potentially valuable or salvageable entries. Therefore, many of the algorithmic approaches employed were designed to accommodate complicated cases and missing data, with the aim of maximizing the number of usable final results. Quality control metrics were captured at the time of processing to support post-algorithm adjudication, back-tracking, and identification of values requiring removal prior to metric calculation.

Specific workarounds are described in each relevant methods section, with brief assessments of algorithmic performance reported in the respective Results section. The scope of **Step 2** was extensive enough that its findings have been compiled in a separate document dedicated solely to reporting these results. Only key insights and conclusions are presented here.

Special Considerations for PO Boxes and Missing `address_line_1` Entries

The raw data review of the 2023 and 2026 Formats revealed several notable special cases requiring specific handling. These are described in the respective sections of Step 2: Clean and Validate Addresses and Geolocation, and Annotate Records with Census Boundaries. Notably, approximately 14% of businesses had at some point filed under a PO Box, and approximately 3% had at some point submitted a record with no `address_line_1` value (`address_line_1 == NA`; no blank strings were observed). For context, each entry in the dataset represents a unique combination of address and year over 1,080,764 unique ABI, with no year repeated within a given business.

Later in the analysis, religious organizations are classified as closed if they did not file for four consecutive years during the observed period; organizations that subsequently resumed filing are classified as reopened. Closures are further differentiated into two categories: those that stopped operations entirely and closures due to relocation. Relocated organizations remained operational but moved outside the community, thereby resulting in a loss of social benefits to

that community.

Addresses filed under a PO Box or missing an `address_line_1` value present challenges for generating these metrics. Dr. Insong's method addressed this by removing all PO Box entries while retaining the remaining addresses for each business. However, since reported dates were unique across entries, this was expected to skew results. The Summer 2025 prototype methods sought to associate each PO Box with a nearby physical address (both spatially and temporally) to serve as a proxy for the organization's location on those dates. This relied on the assumption that PO Boxes are opened near the business's operating location, which was later deemed too strong an assumption to adopt.

For these reasons, under the following three conditions, all entries associated with a business are removed:

- Geocoordinates are absent and could not be recovered through address-based validation, preventing spatial joining (0.39% missing before **Step 2** and 0.03% missing after validation/cleaning).
- `address_line_1` is missing, preventing verification that the geocoordinates correspond to the institution itself rather than a general location within that ZIP code (3% ABI).
- PO Box used: the physical location of the business, and consequently its presence or absence within the community of impact, is ambiguous (14% ABI).

Table 1 shows the distribution of ABIs impacted by any one of these conditions. Not shown are results indicating that longitude and latitude were always missing together, affecting 0.4% of ABIs. These results demonstrate very little overlap between conditions, suggesting that approximately 17% of ABIs, nearly one-fifth of ABI, could potentially qualify for removal.

Table 1: Percentages of ABI with Address and Geolocation Considerations

NA Address = FALSE

	PO Box	
NA Lon	**FALSE**	**TRUE**
FALSE	83.51	13.48
TRUE	0.36	0.02

NA Address = TRUE

	PO Box	
NA Lon	**FALSE**	**TRUE**
FALSE	2.38	0.24
TRUE	0.02	0.00

Simply removing ABIs with these characteristics introduces a potential source of bias, especially among those filing under a PO Box. Table 2 summarizes the share of ABIs affected by PO Box filings and missing `address_line_1` across U.S. cities, with some reaching 100% of ABIs impacted. Figure 1 maps these values spatially, confirming that some communities are

disproportionately affected by this data loss. Note that 10.6% and 12.6% of cities could not be matched to TIGER/Line Shapefile features imported via the `tigris` package.

Table 2: Summary Statistics of ABIs with Select Characteristics by City

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
NA address line 1 (%)	0.09	2.56	6.25	14.93	17.65	100.00
PO Box (%)	0.26	15.38	33.33	42.16	63.64	100.00

The agreed-upon mitigation strategy is to remove an ABI only when the selected date range includes an address with a missing `address_line_1`, a PO Box filing, or an address without geocoordinates and, consequently, no census boundary mapping. The number of ABIs removed under each condition will be reported on the dashboard to inform users of the extent of data loss attributable to each condition.

Step 1: Nomenclature Standardization and Parquet Conversion

Computational Dependencies and Custom Functions

All packages required for this step are documented in the **SET UP THE ENVIRONMENT** section of "Clean Raw Data_Step 1_2026 Format.R". Custom functions used throughout the script are in "Support Functions/For Step 1_2026 Format.R".

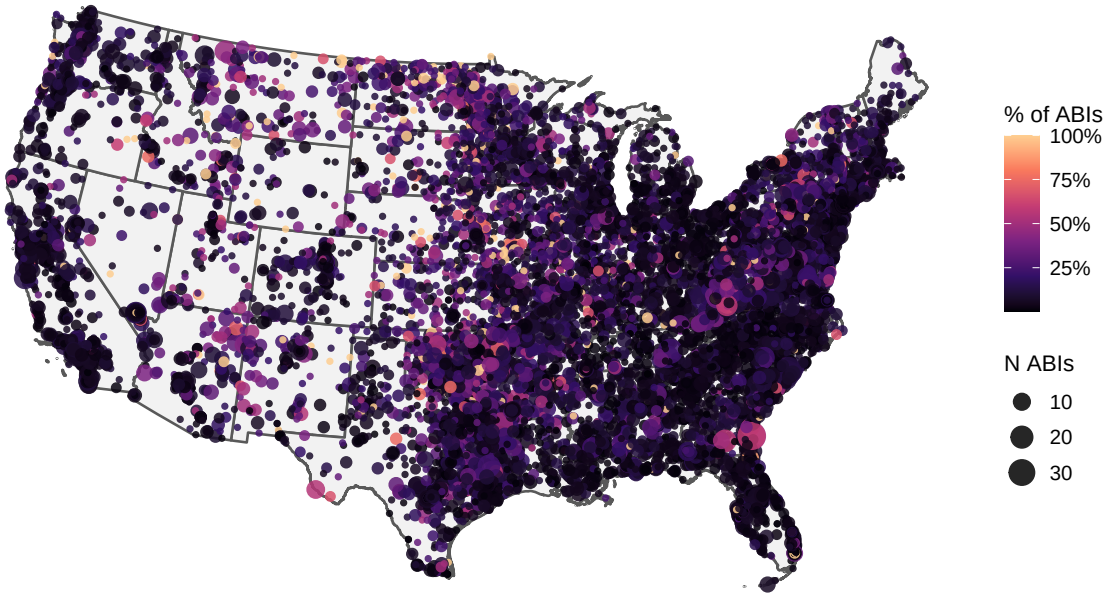
The 2026 Formatted data included a number of additional metadata fields; some were directly relevant to the analysis done here, while many others were not directly relevant or were undefinable. Further details can be found in the "Review_2026 Format.pdf" report. It was decided to retain all metadata fields in accordance with good data science practices, while limiting further validation and processing steps to only those fields relevant to the immediate analysis.

In **Step 2**, address, geolocation, and census boundary variables are thoroughly processed through sequential validation and cleaning measures. Prior to this step, a few additional metadata fields were reformatted but not validated, as the data provider is treated as the known authoritative source of these categorical attributes.

The North American Industry Classification System (NAICS) code is typically six digits long; however, the data provider appended an additional proprietary two-digit code whose encoding

ABIs with Any Missing Address Line 1 by City

Color = % of ABIs; size = N ABIs



ABIs using PO Boxes by City

Color = % of ABIs; size = N ABIs

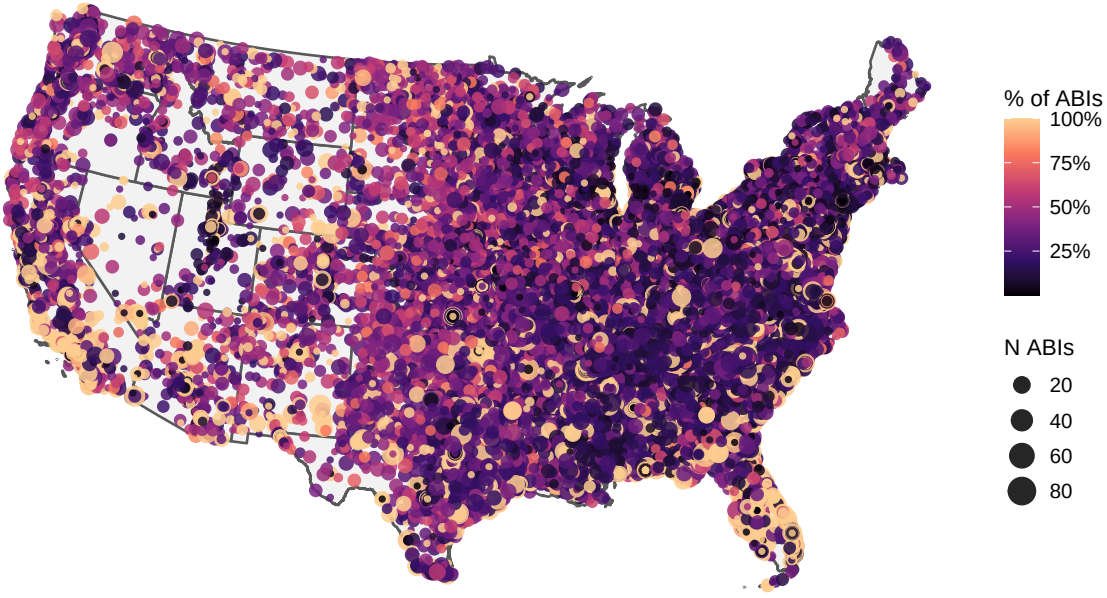


Figure 1

is unknown. These were separated into two columns to clarify the association between the NAICS code and its description, distinct from the additional two-digit code. This yielded three new columns: `primary_naics6_code`, `naics6_descriptions`, and `primary_naics2_code`, the last of which has no associated description.

The SIC codes appear in a primary column, limited to 42 unique values, and four additional overflow columns representing over 2,500 unique attributes, such as denomination, functional capacities (e.g., screen printing, concert venues, stables), and services (e.g., clothing, food banks, substance abuse support). Three assumptions associated with these columns were validated: (1) the additional SIC code columns are overflow columns, where preceding columns are non-`NA`, indicating progressive filling; (2) codes and descriptions are mapped one-to-one; and (3) codes are always associated with a non-`NA` description.

Assumptions were tested using the custom function `sic_overflow_audit()`, which generates all relevant diagnostics. Minor inconsistencies between descriptions associated with the same SIC code were manually revised across all SIC columns using the custom function `clean_sic_descs()`. Any SIC code sometimes associated with a description and sometimes `NA` was revised to use the available description using the custom function `set_sic_desc_for_code()`. If multiple descriptions were associated with the same code, one description was chosen at random using the custom function `clean_sic_codes()`.

As a final step, the dataset was converted from the CSV output of the Data Axle database to Parquet format using the `arrow` package [3]. This columnar format stores large datasets more efficiently, enabling faster read and write operations. It also supports database-style querying, allowing users to limit the amount of data loaded into memory at any given time and making better use of available RAM.

Additional details regarding the algorithm's performance outcomes can be found in Results - Step 1: Nomenclature Standardization and Parquet Conversion.

Step 2: Clean and Validate Addresses and Geolocation, and Annotate Records with Census Boundaries

Computational Dependencies and Custom Functions

All packages required for this step are documented in the **SET UP THE ENVIRONMENT** section of "Clean Raw Data_Step 2_2026 Format.R". Custom functions used throughout the script are in "Support Functions/For Step 2_2026 Format.R".

The algorithm implementing the described methods has been separated into a dedicated HPC script, contained in "Clean Raw Data_Step 2 HPC v2_2026 Format.R", which includes its own **SET UP THE ENVIRONMENT** section. **PART A: UTILIZING THE HPC** further describes how to reproducibly run the script within the HPC environment as both a live session or batch array. The preconfigured shell script used for deploying batch arrays is "Validation SLURM_2026 Format.sh".

Note that the comparative HPC script "Clean Raw Data_Step 2 HPC v1_2026 Format.R" reflects the initial approach assuming unrestricted access to the USPS API. However, in Spring 2026, the USPS announced an update from version 3.2.3 to 3.3.1 with revised terms of use and a new pricing tier that took effect August 1st, 2026. In response, the algorithm was modified to more flexibly support agnostic address compression based on string similarity, and to control API usage via a boolean flag indicating whether the API should be used and, if `TRUE`, a query limit specifying how many addresses were to be validated in that run.

The methods are comparable between the two versions, with the logical flow of the address validation and compression algorithm being the principal difference. As a consequence of the API terms of use update, v2 was used to process the entire dataset. Users should therefore treat v1 as archived for posterity and v2 as the version to be used going forward. The specifics of v1 are not discussed here.

With access to the HPC, all sequential steps that validate and compress addresses, use them for address-based geocoding, and annotate unique ABI and address entries with census boundaries are completed within a single algorithm. Each macro component is prefaced with **LOOP**, corresponding to the respective section of the HPC script.

Each validation section requires user-specified configurations or pre-analysis setup. The specifics of these are described in each relevant section, but must be completed prior to running the algorithm to ensure it operates correctly and within the constraints of the project's available

resources.

LOOP PART A: Isolate Unique Candidate Addresses

Addresses associated with each individual ABI are processed simultaneously. First, each ABI's unique address is identified and saved into `candidate_addresses`, which retains the key address fields. For each unique address, the earliest and latest years it appears are recorded, the number of entries with valid longitude/latitude is counted, and the average longitude and latitude (among valid entries) is computed. Additionally, the boolean `do_api`, indicating whether the USPS API should be used, is carried forward when present. These candidate addresses are then passed through all subsequent processing sections.

If `address_line_1` is missing, the entry cannot be processed for address validation or address-based geocoding. As noted earlier, any given geocoordinates for these records cannot be assumed to represent the business's physical location. Although the algorithm can still annotate these entries with census boundaries, this does not mean they are suitable for mapping closure results. To prevent errors when `address_line_1 == NA`, these records are separated from those with an `address_line_1` so they can be validated and then condensed.

> **!** Note on Geocoordinate Accuracy for Entries Missing the Address
>
> While entries lacking an `address_line_1` but have a valid geocoordinates are annotated with census boundaries, this does not imply that the geocoordinates accurately reflect the physical location of the business. This measure is intended solely to allow the algorithm to run without error, and affords users the opportunity to adjudicate these entries at a later stage if desired.

LOOP PART B: Consolidate and Verify the Addresses

Address heterogeneity arising from typographical errors was observed in the 2023 Format and is assumed to exist in the 2026 Format. In the 2023 version of this stepwise process, addresses were first compressed based on similarity before validation. This ordering was necessary because the workflow was run locally, and reducing the number of unique addresses to validate helped expedite the USPS API step.

However, a post-assessment of the algorithm's outputs showed that the original compression step induced duplications in years-open and years-closed measures. This largely occurred because the same `address_line_1` was reported with differing metadata (e.g., city or ZIP code). After validation, a secondary address-compression step was run on qualifying entries where this issue was detected, using the corrected, standardized address information to recompress records.

With access to the HPC, the compute constraints of address verification at scale became tractable. The revised workflow reflected here resolves heterogeneity in two stages: (1) USPS API verification, which standardizes addresses and reconciles unverified records to their closest verified match using string similarity; and (2) agnostic clustering (a fallback when verification is unavailable or fails), which groups similar addresses by string similarity and resolves conflicts within each cluster to reduce redundant entries.

The same algorithm is used at both points where address string similarity is evaluated, but the way "best" matches are triaged differs between the two stages. To prevent the table from expanding when multiple best matches are identified, a single match is forcibly selected by ranking candidates according to a predefined set of criteria and choosing the top-ranked result.

LOOP PART B.i.: Validate Addresses with USPS Database

Excessive deviation from standard address formatting has the potential to contribute to failed matches with the U.S. Census Bureau's Geocoder API, used in **LOOP PART C** to validate the geocoordinates associated with each address [4]. To mitigate this risk, addresses were first validated against the USPS Address API to retrieve the preferred, standardized form of each address prior to geocoding [5].

> **Alternative Validation Resources**
>
> These APIs were chosen primarily for their relative simplicity of setup and use, their suitability for automation and parallelization within the HPC environment, and their lack of associated cost. However, as described in their respective results sections, these processes were unable to validate as many records as anticipated. Furthermore, as they are separate systems, they required sequential processing, whereas alternative resources can offer both address and geolocation validation within a single query. The following alternative databases were considered for validating the listed physical

addresses and associated geolocations:

- Environmental Systems Research Institute, Inc. (Esri) StreetMap Premium, including the locally hosted version available to Yale affiliates
- Google Maps API

These sources were ultimately not used due to a combination of limiting factors. Under the applicable Data Use Agreements (DUAs), it was necessary to confirm that no API usage would violate the terms of those agreements. The USPS and U.S. Census Bureau APIs were approved following coordination with the relevant security review processes; however, accessing the Esri online database or Google Maps API required additional review.

The review process was initiated for the Google Maps API given its anticipated capacity to efficiently validate addresses and geolocation. However, the review extended beyond the project timeline, rendering those resources inaccessible. Additionally, Esri requires manual adjudication of alternative address suggestions, which would have been prohibitively time-intensive. Researchers from the Yale Center for Geospatial Solutions (YCGS) further noted that the online Esri resource is more current and robust than its locally hosted counterpart, suggesting that the local version would offer little advantage over the USPS Address database in terms of address validation quality.

⚠ USPS API Version 3.3.1 Update: Compatibility and Pricing

As of August 1st, 2026, the USPS upgraded the API from version 3.2.3 to 3.3.1 and updated their terms of use to include tiered pricing for API access.

Users should be aware that the algorithm reported here was designed for version 3.2.3 and may not be fully compatible with the updated API without modification. Additionally, the 2023 Format implementation assumes unrestricted access to the API, whereas the 2026 Format implementation was updated with a version 2 allowing users to specify the number of addresses queried against the USPS API, enabling cost management.

In response to the updated terms of use, the workflow was revised to give users more control over address verification. All settings are outlined and explained in **SUBSECTION B3: Optional USPS Address Verification (API Setup and Quota Controls)**. Users can set the boolean `verify_addresses` to indicate whether verification should be run, and API credentials are imported only when `verify_addresses == TRUE`.

Users can also define an address-count quota via `set_quota`. This quota is then applied through an if/else selection and custom function `usps_quota_mark_api()` to determine which ABIs are sent for verification. Selection is made at the ABI level: if an ABI is chosen, all of its addresses are verified, and no partial ABI selections are allowed. As a result, `set_quota` functions as an upper bound on the number of addresses verified (the total may fall below the quota to avoid splitting an ABI). The if/else conditionals generate a message indicating whether the API will be used, how many ABI entries will be validated, and whether addresses were randomly sampled. A set seed allows users to reproduce any randomized selections.

When `verify_addresses == TRUE` the algorithm queries the USPS Address API using the API specifications, methods, and sample processes outlined in the official developer documentation and the USPS API Examples GitHub repository [5,6]. Pre-configured customer key and secret credentials are used to generate an API token for each address query via the custom function `generate_usps_token()`. The raw address fields, address lines 1 and 2, city, state, and five-digit ZIP code, are submitted to the database, and the response JSON is parsed to extract validated address components into discrete columns via the custom function `validate_usps_address()`. Both functions utilize the `httr` and `jsonlite` packages for API interaction [7,8].

If the initial USPS validation did not return a match, one fallback strategy was applied using the recommended city associated with the five-digit ZIP code. If this also failed to produce a match, the query returned "No address match found".

This fallback was particularly important for the 2023 Formatted dataset. Initial assessment showed that ZIP code fields had been converted to numeric values at some point, stripping leading and trailing zeros. Before querying, ZIP codes were normalized back to the expected five-digit format, but any padded zeros were added arbitrarily. Although the 2026 Formatted dataset correctly preserves ZIP codes as character variables, this step was retained as a precaution in cases where the provided city does not align with the ZIP code and ZIP codes got mistakenly converted to numeric.

In the **LOAD IN THE DATA** section, users can indicate whether ZIP codes were retained as character strings (i.e., no flanking zeros were stripped) by setting the boolean `zip_codes_character`. When `zip_codes_character == TRUE`, the fallback uses the custom function `get_city_info()` to retrieve the recommended city for a given ZIP code. The resulting valid ZIP code–city pair is then substituted for the raw values and the USPS query is resubmitted. The ZIP–city lookup table is built from the SimpleMaps Basic United States Cities Database (version 1.90) using the custom function `build_zip_city_lookup()` [9].

If ZIP codes are indicated as having lost flanking zeros, candidate ZIP code variants are first generated using the custom function `make_zip5_candidates()`. For example,

`make_zip5_candidates("01230")` would generate “01230”, “00123”, and “12300”. The custom function `get_city_info()` then retrieves the recommended city associated with each candidate ZIP code. Valid ZIP code–city pairings are subsequently substituted for the raw values to resubmit the query, though the algorithm attempts only the first valid pairing returned by `get_city_info()`.

Five quality-control columns were used, including: (1) a boolean indicating whether the address was validated; (2) a character field indicating which attempt succeeded (e.g., “First try” or “With city correction by zip5”); and (3) the USPS API query status. The status codes were decoded using the custom function `usps_status_group()`, based on the expected outcomes defined in the USPS API documentation.

> **API Credentials and Configuration**
>
> Users who wish to utilize this resource for their own work will need to register for a USPS Developer account, sign the license agreement, and configure their own individual API client key and secret.
>
> The client key and secret must be kept private and must not be shared. Instructions for configuring these credentials are provided in the header of “Code/Clean Raw Data_Step 2_2026 Format.R”.

LOOP PARTS B.ii--B.iii: String Similarity Matching for Unverified and Heterogeneous Address

The same underlying algorithm is used whenever string-similarity matching is performed. However, best-match selection differs by phase (matching with unverified addresses vs. “agnostic” matching). In both phases, candidate pairs are evaluated under two match classes: “exact” and “fuzzy.”

Exact matches. Exact matches compare `address_line_1` using threshold $\delta = 0$ and are treated as valid regardless of associated address metadata. In these cases, its assumed the likelihood that the records represent separate addresses due to a true physical move is low. Accordingly, the geolocation test is overridden and the match is retained.

Fuzzy matches. Fuzzy matches compare `address_line_1` using threshold $\delta > 0$ but ≤ 0.2. For these match pairs, the longitude and latitude differences are required to fall within 0.02 degrees each. If the geolocation test fails, the match is rejected and the records are kept

separate.

Verified-Address Matching

This step occurs after addresses are querried against the USPS API. Unverified addresses are triaged by attempting to link each record to a verified address using `address_line_1`.

The algorithm first attempts an exact match by selecting the verified candidate whose years-open interval overlaps (or touches) the record's window, or—if none overlap—the candidate with the smallest year-gap; geolocation checks are skipped. If multiple candidates remain, ties are broken by preferentially selecting the closest match occurring before the window, and then after, then forcing only a single top candidate is retained.

If no exact match exists, the algorithm attempts a fuzzy match using the same years-open selection rule but requiring geolocation validation; records that fail validation remain separate. Fuzzy matching is run iteratively starting at $\delta = 0.2$ and then applying progressively more stringent similarity thresholds until each unmatched address resolves to a single best candidate; if no match is found at any threshold, the record is passed to the next section.

Agnostic Matching

This step runs after addresses are queried against the USPS API and unmatched records remain, or when no address validation is performed (i.e. either `verify_addresses == FALSE`, or `verify_addresses == TRUE` but `all(unver_candidates$do_api == FALSE)`). Because there is no predefined pool of verified candidates, the algorithm selects best matches agnostically from the addresses available within each cluster.

Within each exact `address_line_1` group, matching is restricted to records sharing the same expected city for the ZIP code, as defined by the lookup table produced by `build_zip_city_lookup()`. Using the custom function `pick_best_by_city()`, rows are split into trusted anchors where the reported and expected city are the same (`city_match == TRUE`) and conflicting records (`city_match == FALSE/NA`). If no trusted anchor exists, the best conflict row is promoted to serve as an anchor. Each record is then linked to a single best anchor, prioritized by (1) years-open overlap (or, if none overlaps, the smallest year-gap), (2) presence of geolocation, (3) smallest lon/lat difference across candidates, and (4) a stable tie-break. The geolocation test is bypassed and recorded as an override. Records with no anchor in their city are left for the fuzzy pass.

For fuzzy matching, similar strings are clustered and one anchor per validated city is chosen.

Anchors are preferred in order: presence of an exact match from the prior step, earliest year recorded open, then a stable tie-break; if no prior match exists, the "best" anchor is chosen based on geolocation pass rate, mean year gap, and geolocation coverage. Candidate pairs are rejected when both sides have coordinates and either longitude or latitude differences exceed 0.02 degrees. If either record in the pair is missing coordinates, the missing values are ignored and the test is recorded as `"PASS; no Lon"`, `"PASS; no Lat"`, or `"PASS; no Lon/Lat"` to indicate which geocoordinate is absent. Fuzzy matches are applied only when `geo_valid == TRUE`; otherwise records remain unmatched, with provenance recorded as `address_matched = "Fuzzy match"` and `match_geolocation_test` indicating whether the geolocation check was evaluated/passed.

For fuzzy matching, similar strings are clustered with $\delta = 0.2$ and one anchor per validated city is chosen, first preferring any exact matches created in the prior section. If no exact match is available, the anchor is selected based on how well each candidate represents the rest of the pool using the custom function `pick_best()`. This algorithm chooses the record whose coordinates agree with the largest number of other records (within the longitude/latitude difference < 0.02 degrees) and, if needed, the record whose years-open range overlaps with the most others. If ambiguity remains, it selects the record with the smallest overall year separation from the remaining records, then favors entries with a ZIP+4 and a larger number of records with valid geocoordinates for that address, and finally applies a stable alphabetical tie-break on the combined address string.

Fuzzy Matching Algorithm

Different uniquely listed address line 1 (`address_line_1`) strings for a given business (`abi`) were fuzzy matched using the Jaro-Winkler method available through `stringdist(method = "jw")` in the custom function `find_similar_addresses()` [10]. This edit distance method measures string similarity by accounting for the proportion of characters shared between two strings and the number of transpositions needed to place matching characters in the same order, normalized by string length [11–13]. Scores range from zero to one, where a score of zero indicates identical strings and smaller values reflect greater similarity.

Prior to processing, addresses were normalized in the custom function `preprocess_address()` by converting all characters to lowercase, normalizing spacing and punctuation, removing non-alphanumeric characters (with the exception of commas and spaces), and trimming extraneous whitespace.

${}_nC_2$ pairwise comparisons between n unique normalized `address_line_1` strings produce a symmetric $n \times n$ string distance matrix, that is $d(i, j) = d(j, i)$ and the diagonal entries

$d(i, i) = 0$ represent the trivial self-similarity solution. To reduce redundancy, the algorithm evaluates only one orientation of each pair and omits self-similarity comparisons, computing only the upper triangle of the matrix. Addresses with a pairwise Jaro-Winkler score below a set threshold, δ, are saved as a qualifying match. The matching threshold $\delta > 0$ and ≤ 0.2 was determined empirically through testing on a sample of addresses, confirming sufficiently accurate grouping of similar entries without the erroneous inclusion of dissimilar ones.

Computational Performance

The time complexity of this operation is $\mathcal{O}\left(n^2 \times l\right)$, where n is the number of unique addresses and l denotes the computational cost of the Jaro-Winkler algorithm for a string of length l [13–15].

Because the algorithm operates solely on `address_line_1`entries, string lengths remain relatively short, making the contribution of l negligible. Additionally, because the number of unique addresses is expected to be bounded by the number of dates in the observed period, the computational demand of this method is not expected to be a limiting factor in the algorithm.

Consider the following example of a business that filed under two distinct addresses, each of which was reported with slight variations, resulting in five unique address strings.

Node	Address
1	`123 Main Street`
2	`123 Main St`
3	`123 Main St.`
4	`456 Oak Avenue`
5	`456 Oak Ave`

Using `stringdist(method = "jw")` the pairwise distances are computed for all ${}_5C_2 = 10$ unique address pairs. This yields the following upper-triangular distance matrix:

	1	2	3	4	5
1	—	0.08	0.09	0.41	0.38
2		—	0.05	0.39	0.35
3			—	0.40	0.36
4				—	0.11
5					—

Version: 006.07.15.26
Generated using Quarto (v. 1.8.27)
and RStudio (v. 2026.01.1+403)

Pairs falling below the threshold $d(i, j) \leq \delta$ are considered candidate matches and recorded as edges in an undirected graph, where each node represents a unique address. This graph is stored as an adjacency list, like the one shown below [16]:

Node	Address	Nearest Neighbors	JW Distances
1	`123 Main Street`	$\{2, 3\}$	$\{0.08, 0.09\}$
2	`123 Main St`	$\{1, 3\}$	$\{0.08, 0.05\}$
3	`123 Main St.`	$\{1, 2\}$	$\{0.09, 0.05\}$
4	`456 Oak Avenue`	$\{5\}$	$\{0.11\}$
5	`456 Oak Ave`	$\{4\}$	$\{0.11\}$

A Depth-First Search (DFS) algorithm in the custom function `find_components()` is applied to the adjacency list to identify connected components, where each component forms a cluster of similar addresses.

Algorithm 1 Depth-First Search — Connected Component Identification

Require: Adjacency list `address_graph`, node set $\{1, \ldots, n\}$

Initialize an empty list of clusters
for each node i in $\{1, \ldots, n\}$ **do**
    Begin a new traversal from node i
    Initialize an empty cluster
    repeat
        Add the current node to the current cluster
        if the current node has an unvisited neighbor **then**
            Move to an unvisited neighbor and continue
        else
            Backtrack to the previous node
        end if
    until no unvisited neighbors can be reached in the current traversal
    Record the cluster
end for
Remove duplicate clusters
return list of unique clusters

One candidate address is selected from each cluster using the triage rules described above to serve as the best match and anchor for associating the other addresses in the pool. Because the reliability of string-based matching can vary with string length, an additional geographic validation step is applied: for each matched set, the absolute range in reported longitude and

latitude is computed as the difference between the maximum and minimum values.

For reference, one degree of longitude is approximately 69 miles (111 kilometers), while one degree of latitude varies with proximity to the equator, averaging approximately 54 miles (87 kilometers) across the contiguous United States [17,18]. The length of a typical U.S. city block similarly varies, with common estimates ranging from 100 to 200 meters [19,20]. Based on these benchmarks, address sets in which both the longitudinal and latitudinal differences exceeded 0.02 degrees ($\approx$ 2 kilometers or $\approx$ 1.4 miles) were flagged as invalid matches and retained as distinct addresses in the final dataset.

This approach assumes that reported geolocations for each address are accurate to within the specified range with sufficient consistency to be reliable. As the primary objective of this method is to reduce address reduplication and improve downstream algorithmic performance, variations that may undermine this validation step are not considered to meaningfully impact the final results. When uncertain, the method is designed to err on the side of caution by retaining addresses as distinct rather than risk consolidating genuinely separate addresses that share strong surface-level similarities.

LOOP PART C: Verify Geolocation with the US Census Bureau's Geocoder Database

The dashboard visualizes aggregated addresses at multiple census geographies—block group, tract, county, state, and ZIP Code Tabulation Areas (ZCTAs). While small location differences have little impact on county-level summaries, they can substantially affect assignment to finer boundaries such as block groups and tracts.

In the 2023 Format, results from "Step 1: Address Deduplication via String Similarity Matching" showed that geocoordinates for records sharing the same `address_line_1` (and thus potentially the same physical location) sometimes differed by as much as 3 degrees. At mid-latitudes, a 3-degree difference is roughly 150–175 miles—about the distance between New Haven, CT and Philadelphia, PA.

To mitigate these errors, addresses are geocoded using the U.S. Census Bureau's Geocoder API to retrieve the coordinates assigned to each address, prioritizing verified addresses when available [4]. The API's documentation does not specify which part of the building the coordinates represent, such as the delivery entrance or rooftop centroid [21]. All available address fields (address line 1, city, state, and five-digit ZIP code) were used to find matches, prioritizing verified addresses over raw, given values.

Raw address fields are submitted to the Census Geocoder API, and the response

JSON is parsed to extract validated address components into discrete columns via the custom function `validate_geocoordinates_geoCoder()`. This function internally calls `call_census_geocoder()`, which relies on the `httr` and `jsonlite` packages for API interaction [7,8]. The geocoding process first builds an ordered list of benchmark/vintage candidates, then iterates through them in priority order until a successful match is found.

In **SUBSECTION B2: Set Geocoder Search Priorities**, users specify the ordered list of benchmark/vintage pairs the geocoder should try. If users prefer not to use the default triage list, **SUBSECTION A2: Set Geocoder Search Priorities** in "Clean Raw Data_Step 2_2026 Format.R" provides the custom function `census_geo_show_options()`, which retrieves all available benchmark and vintage options that can be used for configuration.

RStudio Console

```
# Construct a prioritized table of benchmark/vintage search criteria,
# where earlier rows take precedence.
spec <- tibble::tibble(
   benchmark_name = c("Public_AR_Census2020", "Public_AR_Census2020",
                      "Public_AR_Current",  "Public_AR_ACS2025"),
   vintage_name   = c("Census2020_Census2020", "Census2010_Census2020",
                      "Current_Current",       "Current_ACS2025")
)

# Construct the "tries" input for `validate_geolocation()`
geocoder_census_tries <- census_geo_make_tries(spec)
```

For each candidate, the vintage is resolved to a numeric ID using `resolve_vintage_id()` — leveraging a per-run cache and skipping any candidates that fail due to lookup, network, HTTP, or parsing errors — before constructing a request URL, calling the API, and logging the outcome. An example query URL constructed from parsed address components using the custom function `build_addr_geo_url()` is shown below.

RStudio Console

```
httr::modify_url(
    "https://geocoding.geo.census.gov/geocoder/geographies/address",
    query = list(
```

```
        format    = "json",
        benchmark = benchmark,
        vintage   = vintage,
        street    = street,
        city      = city,
        state     = state,
        zip       = zip
      )
    )
```

The API sometimes returns multiple possible matches. If there are multiple candidates, the custom function `select_best_match()` first tries to resolve the choice using the ZIP code by preferring a candidate whose ZIP uniquely matches the input ZIP (treating ZIP+4 as the 5-digit ZIP). If ZIP does not identify a single candidate, it compares the input address text to the candidates' matched-address text using `find_similar_addresses()`, progressively tightening $\delta = 0.2$ until one candidate is most clearly paired with the input. If that still does not produce a clear winner, it defaults to the first candidate in the remaining list.

LOOP PART D: Point-in-Polygon Spatial Assignment of Census Information

Refer to Appendix - GEOID Prebuild Shapefiles, Core Area CBSA/CSA Prebuild Shapefiles, Core Area ZCTA Prebuild Shapefiles for information on where to find the code that generates these results, as well as the results codebook.

The data spans three decennial years: 2000, 2010, and 2020. Because users can select varying date ranges and may expect temporally accurate census boundary mappings, each address must be annotated with the census boundaries associated with all three. Once annotated, entries can be aggregated to the census boundary identifier from a given decennial US Census Bureau's TIGER/Line Shapefile release [22]. This enables results to be spatially projected onto maps via the decennially accurate demarcations of census boundaries.

Accurate spatial depiction of these trends across census boundary levels requires consideration of varying degrees of locational error. Counties and states represent larger areas, making them less sensitive to imprecise locational information and less prone to boundary changes between decennial periods. In contrast, smaller boundaries such as block groups and tracts can change substantially between decennial periods and are more sensitive to locational errors. Ensuring accurate fine-level annotations for historical decennial years is therefore an important consideration.

The preceding sections focus on validating and improving the geocoordinates associated with each business through address-based geocoding. When attaching census boundary identifiers to these records, there are two fundamental approaches:

1. Referencing a database with decennial boundary information already associated with each address.

2. Assigning decennial boundaries to each address via point-in-polygon spatial joining of verified longitude and latitude coordinates against reliable spatial representations of map boundaries, known as polygons.

The second approach is the standard of practice, ensuring all desired decennial boundary information is correctly assigned to the provided geocoordinates. Its accuracy is limited only by the reliability of the available address coordinates and boundary shapefiles. Figure 2 demonstrates how geocoordinate points are interpreted by a spatial join to polygon geometries, such as county-level census boundaries, geographically.

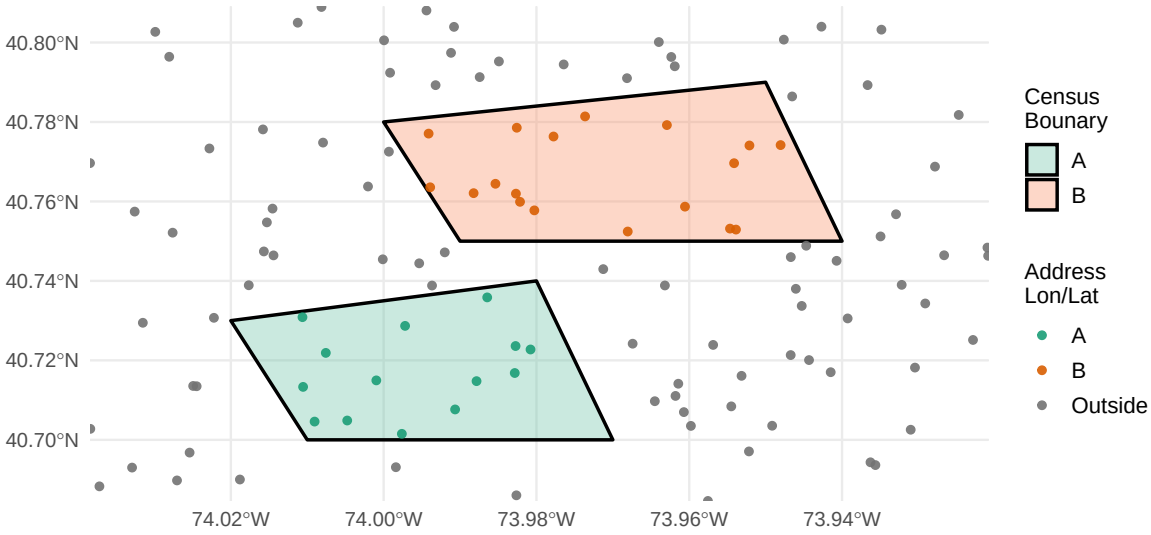


Figure 2: Illustration of Point-in-Polygon Spatial Joining

Computational Performance

The time complexity of a naïve spatial join for a single decennial period is $\mathcal{O}\left(N \times M\right)$, where N denotes the number of points and M the number of polygons [14,23].

This can create significant time lags when associating multiple addresses across many boundaries, such as block groups or tracts in a densely populated region. Steps leading up to the point-in-polygon operation reduce the number of comparisons by consolidating addresses and filtering regions down to the state level. The county and other census boundaries were not used for filtering, as they were assumed to vary between decennial periods.

Other strategies exist that can improve the speed of this operation. For example, spatial filtering or indexing can reduce the complexity to $\mathcal{O}\left(N \log(N) \times M \log(M)\right)$ [23]. Due to time constraints developing the process, these were not explored, and some entries, particularly those spanning denser census regions with more addresses, took orders of magnitude longer to compute than others.

While developing the pipeline for the 2023 Format, three sources of census boundary information were explored, all of which involved API calls to retrieve the requisite information at the time of annotation. This process exacerbated what was already a computationally costly spatial join operation, and was expected to worsen once the algorithm was expanded to include block group-level census codes, which are notoriously slow to query. Additionally, the `tigris` process exhibited issues retrieving data for the 2000 and 2020 decennial years.

To improve performance and accuracy in preparation for migration to the HP environment, the relevant TIGER/Line Shapefiles were manually downloaded and preprocessed. Each shapefile was structured to include the desired metadata by decennial year as discrete layers. The pipeline supports both block-level and block group-level shapefiles: both contain polygon boundaries that associate block or block group, tract, county, and state codes with any longitude/latitude coordinate falling within a given polygon. Block group-level shapefiles are preferred for generating summary statistics due to their reduced complexity compared to block-level shapefiles.

In the 2026 Format workflow, two additional geographic codes representing Core Based Statistical Areas were associated using the same method: Metropolitan/Micropolitan Statistical Areas (denoted as CBSA) and Combined Statistical Areas (CSA). These designations were first introduced in 2003, replacing the Office of Management and Budget's (OMB) prior use of "Standard Metropolitan Areas." CBSA and CSA boundaries are typically updated annually, every five years at mid-decade, and following each decennial census. For the purposes of this processing, the 2010 and 2020 decennial releases were used alongside the earliest available annual release from the 2000s decade (the 2007 vintage) to provide coverage across all relevant time periods.

Additionally, ZIP Code Tabulation Areas (ZCTA) national files were obtained for each decennial year and compiled alongside the CBSA/CSA output. To speed up processing, the state acronym is joined during preparation of the compiled core areas file (containing the combined CBSA/CSA and ZCTA decennial details as layers).

Six types of census boundary shapefiles are used: state block-level, state block group-level, national CBSA, national CSA, national ZCTA, and national states (and equivalents). Each was obtained for the 2000, 2010, and 2020 decennial vintages, except CBSA and CSA, for which the 2007 vintage was substituted for 2000. These shapefiles were processed and compiled as GeoPackage (`*.gpkg`) files in `"Data/Results/Census Bureau TIGER Line Shapefiles/"`, with state block- and block group-level data organized as layers by decennial year. CBSA, CSA, and ZCTA data (annotated with the state acronym) were compiled into a single file, with one layer per census boundary type and decennial year combination.

Unlike the CBSA and CSA shapefiles, the ZCTA shapefiles did not contain any state information. This was added via a spatial join with the state-level shapefile for the respective decennial year.

Census Boundary Files: Download and Local Build Requirements

All census boundary files, both the downloaded raw shapefiles and the compiled GeoPackage versions, are too large for effective Git tracking and distribution. New users will need to download the source files independently and generate the GeoPackage files locally. For download instructions and links to sources, refer to the Directory Specific Notes in the project GitHub repository. For the coded implementation, refer to **SUBSECTION A3: Build Precompiled TIGER/Line GeoPackages** in "Clean Raw Data_Step 2_2026 Format.R".

Because this process is involved, a dedicated set of functions was created for the generation of these reference shapefiles: "Support Functions/For Step 2_Compile Decennial Data.R". The custom function `preflight_tiger_outputs()` first assesses the local environment for the status of shapefile downloads and builds, flagging any missing files or incomplete downloads to the user.

In brief, standardized decennial spatial layers for all census boundaries are compiled using `standardize_geo_layer()` for block and block group-level state GEOIDs, and `standardize_core_area_layer()` for CBSAs, CSAs, and ZCTAs. These standardized files are then written using `write_state_gpkg()` for GEOIDs and `write_core_areas_gpkg()` for CBSAs, CSAs, and ZCTAs, following the execution of

their respective `build_cbsa_csa()` and `build_zcta()` functions.

For size considerations, the GEOID files are retained separately by state, while the CBSA, CSA, and ZCTA layers are compiled as nationally representative files.

For each array run, the dataset is organized by state then ABI to mitigate the number of block group-level GeoPackages loaded simultaneously. GeoPackages for each state represented in that subset are imported using the custom function `read_state_gpkgs_for_data()` under **SUBSECTION B4: Load Relevant Block-Level GeoPackages by State**, creating one layer per state where, within each state, there is one layer per decennial period. Users must predefine the geography level — $\in c(\text{"blocks"}, \text{"block group"})$ — using `block_geography` to specify which level to utilize. The algorithm is configured to process either; while the block-level files contain block group information, processing only at the block group level yields faster performance.

The core areas shapefile contains national-level layers for all three areas (CBSA, CSA, and ZCTA) census boundaries across the three decennial periods. Prior to spatial joining, these files are filtered by the `area_states` column, which is extracted from `area_name` during the shapefile building process.

Spatial joining was conducted using the `sf` package via the custom functions `add_decennial_geoid_block()`, `decode_zcta()`, and `decode_cbsa_csa()` [24]. Geocoordinates were converted to the standard World Geodetic System 1984 (WGS84) and projected into the coordinate reference system (CRS) of the census boundary polygons. A generalized outline of the algorithm executed by each function is outlined below.

Algorithm 2 High-Level Procedure: Spatial Join for Census/Area Identifier Assignment (CRS alignment; optional state partitioning)

Require: Points `cand_sf` (`sf` POINT) with unique `row_id` and optional `state`

Require: Polygon layer(s) containing boundary ID field(s) `id_col` (e.g., Census `GEOID`, block `geoid_block`, ZCTA/CBSA/CSA codes)

Ensure: Table keyed by `row_id` with attached ID(s) (often year-suffixed for decennial layers), `NA` if unmatched

1. **Hard-stop checks:** inputs are `sf`; `row_id` exists; `id_col` exists in polygons
2. **CRS match:** `pts` $\leftarrow$ `st_transform(cand_sf, st_crs(poly_sf))`
3. **Keep only needed polygon fields:** `poly_id` $\leftarrow$ `poly_sf[id_col]`
4. **Point-in-polygon join (left join):** `joined` $\leftarrow$ `st_join(pts, poly_id, join = st_within, left = TRUE)`
5. **Return:** drop geometry; output `row_id` and attached boundary IDs (e.g., `geoid_2000`, `geoid_2010`, `geoid_2020`)

   Variations (mapped to functions):

   add_decennial_geoid_block(): loop by **state** $\rightarrow$ loop by **year** (2000/2010/2020); join to each state-year layer; merge year-specific IDs by `row_id`.

   decode_zcta(): use a **national** polygon layer; optionally pre-filter polygons by candidate states via an `area_states` field; then do a single join to return a ZCTA code column.

   decode_cbsa_csa(): optionally pre-filter by `area_states`; split polygons into CBSA vs. CSA; perform **two** joins and return CBSA code/level plus CSA code (year-suffixed).

Addresses where no `address_line_1` was reported were reintroduced to the running candidate address dataset and, if valid geocoordinates were present, annotated with census boundaries. As noted above, this does not imply that the geocoordinates are valid for mapping closures.

Addresses were also annotated to indicate whether sufficient geocoordinates were available for GEOID joining and whether any match was found. Users should note that it is possible for addresses to not match with a core statistical area (CBSA or CSA), as these boundaries largely represent urban areas rather than rural ones.

LOOP PART E-I:

Up to this point, only a subset of the available columns had been evaluated in the preceding steps: address variables, summaries of year-open date ranges, geocoordinates, and all compiled validation results and quality control outcomes. The remainder of the loop rejoins this table with the rest of the metadata, which is then processed through three subsequent quality control assessments designed to generate insights at the time of execution.

These quality control tables also serve as the location where all raw values are presented alongside their relevant validation outcomes and quality control variables. Finally, all excess columns are consolidated into the final dataset form, prioritizing validated results over unvalidated versions and retaining a limited set of quality control columns to facilitate quick adjudication and back-tracking.

Refer to Results - Step 2: Clean and Validate Addresses and Geolocation, and Annotate Records with Census Boundaries for results on the three quality control assessment outcomes. For the implementation of these loop parts, refer to "Code/Clean Raw Data_Step 2 HPC v2_2026 Format.R". The input data consisted of the standardized and converted dataset generated in **Step 1**: `"./Data/Results/KEEP LOCAL/From Clean Raw Data/Step 1_2026 Format/church_2026_form_standardized_06.10.2026.parquet"`. Results were written to `"./Data/Results/KEEP LOCAL/From Clean Raw Data/Step 2_2026 Format/"`.

Compiling Results and Evaluating Import Quality

Results were saved in two columnar formats: Parquet and DuckDB [3,25]. Parquet files are easy to handle during importation and exportation, but can only contain one dataframe at a time, much like a `*.csv` file. In contrast, DuckDB supports saving lists of dataframes, much like an `*.xlsx` file. To support the writing and reading of DuckDB files, custom functions were created for this purpose. Directories containing result Parquets, representing subset results from batch arrays, are compiled using `compile_parquet_folder()`. Directories containing result DuckDB lists are compiled using `compile_duckdb_folder()`.

Both functions also include generalized import quality checks (QC), which use the file name to verify that all expected ABIs are present in a defined subset. This check requires precise alignment between the dataset organization in the HPC script and the range defined in the file name, and allows for a quick confirmation that no information was lost across batch array runs. Additionally, `compile_parquet_folder()` generates summaries of the discrete QC outcomes (address validation and matching, census annotation, and geocoordinates) saved in the main

dataset alongside all results and metadata.

Two batches were run to complete the process described here. Any subsetting of the ABI search space, which defines which ABIs are processed during a given batch, is described in each respective section in **PART C: Recompile Results from the HPC**. Some QCs were also revised for conciseness and to account for variations not considered or covered by the HPC script. This was done in preparation for compiling the results across batches into the final result in **SUBSECTION C3: Combining the Batches**.

Primary datasets, comprising the main data with all metadata and quality control data, were compiled across batches, while import quality controls were maintained as separate lists for each batch. Batch-specific results are denoted by the first two digits of the batch number it is from. The main dataset was too large to compile in-memory, necessitating a DuckDB SQL compilation approach (code below). Results were then consolidated using the custom function `write_qc_groups()`, combining the compiled main data with their respective batch import quality control datasets.

RStudio Console

```
## SUBSECTION C3: Combining the Batches
# ~ line 1089-1141

# Open a writable DuckDB connection to the target database file.
con <- dbConnect(duckdb::duckdb(), dbdir = out_db, read_only = FALSE)

# Build/refresh "data" by unioning parquet batches (align by name) and
↪  sorting
# by ABI then archive_version_year (asc; NULLs last).
dbExecute(con, sprintf("
    CREATE OR REPLACE TABLE data AS
    SELECT *
    FROM (
    SELECT * FROM read_parquet('%s/**/*.parquet')
    UNION ALL BY NAME
    SELECT * FROM read_parquet('%s/**/*.parquet')
    )
    ORDER BY
    abi ASC,
    archive_version_year ASC NULLS LAST;
    ", p18, p20))
```

```r
# Write QC outputs as separate tables named qc_df_18__* and qc_df_20__*
# (skips NULLs without templates).
write_qc_groups(
  con,
  list(
    qc_df_18 = qc_df_18$qc,
    qc_df_20 = qc_df_20$qc
  ),
  prefixes = c(qc_df_18 = "import_qc_18", qc_df_20 = "import_qc_20"),
  on_missing = "skip",
  verbose    = TRUE
)

# Closes the connection and persists changes to disk.
dbDisconnect(con, shutdown = TRUE)
```

The three at-point-of-analysis quality control datasets were smaller and easier to compile in memory. All outputs were saved using `write_list_to_duckdb()`, which also consolidated all datasets alongside their respective batch import quality control datasets.

For the implementation of data variable assessments relevant to the cleaning algorithm, post-evaluation importation, compiling, and checking, refer to "Code/Clean Raw Data_Step 2_2026 Format.R". The input data consisted of the standardized and converted dataset generated in **Step 1**: `"./Data/Results/KEEP LOCAL/From Clean Raw Data/Step 1_2026 Format/church_2026_form_standardized_06.10.2026.parquet"`. Results were written to `"./Data/Results/KEEP LOCAL/From Clean Raw Data/Step 2_2026 Format/"`.

Additional details regarding the algorithm's performance outcomes can be found in Results - Step 2: Clean and Validate Addresses and Geolocation, and Annotate Records with Census Boundaries, which includes results specifically assessing the outcomes of the three quality control assessments.

Step 3: Standardize and Reshape Metadata to Wide Format

At this stage, the cleaned and validated data is prepared for dashboard metric generation. A critical structural transformation required for this involves reshaping records from long format (years open represented as rows) to wide format (years-open/years-closed encoded as binary columns), such that each record is unique to a given ABI and address combination.

As demonstrated in **PART C: HANDLING ADDITIONAL METADATA** of "Process Data Update.R", many metadata fields are not unique to a given ABI/address combination. While a consolidation procedure was defined that would preserve these variables as strings retaining the unique years reported per ABI/address, they provide no analytical value at this stage of the project. Furthermore, converting them to wide format would render them largely unsuitable for usage.

Given that these variables are outside the scope of cleaning and validation and do not contribute to subsequent steps, they will be excluded from further processing. Users requiring distributor-provided metadata should use the cleaned and validated long-form 2026 Formatted dataset generated after **Step 2**. In-scope entries will be normalized to be unique within each ABI/address combination.

Retained variables: `archive_version_year`, `abi`, `address`, `address_verified`, `address_matched`, `primary_sic_code`, `sic_code[_1:4]`, `sic6_descriptions_sic[1:4]]`, `latitude`, `longitude`, `geolocation_verified`, `geoid_[2000|2010|2020]`, `geoid_match`, `cbsa_code_[2007|2010|2020]`, `cbsa_level_[2007|2010|2020]`, `csa_code_[2007|2010|2020]`, `zcta_[2000|2010|2020]`

Normalize SIC Codes and Annotate with Classifiers

"SUBSECTION B2: Primary and Additional SIC Encodings" in "Process Data Update.R" evaluated the dimensionality, nomenclature consistency, and other characteristics of the SIC code columns. The primary SIC column was found to be restricted to a limited set of codes, while the remaining columns represented thousands of additional functional classifications. Each non-primary SIC column represented a sequential overflow of classification information. Notably, while the primary column contained only 42 distinct codes, those same codes appeared across all overflow columns as well.

Without direct assessment, it is assumed that these metadata columns, particularly following the data cleaning and validation steps that reconciled similar addresses, are not consistent across unique ABI/address combinations. To support downstream annotation, these columns were standardized to ensure a unique value per ABI/address combination, consistent with all other retained variables.

Because the codes in the primary SIC column are restricted but are also represented across the SIC overflow columns, they are handled separately. First, the primary SIC column is normalized by ABI/address combination, with any extra classifications reassigned to the overflow SIC columns. Next, all unique SIC classifications across the overflow columns are pooled together,

including any additional primary SIC codes, and cast to new columns for each additional classification. All primary SIC codes were non-missing and did not need to be supplemented from the overflow columns.

Assign Religious Classifiers

As noted earlier, each ABI and address entry has at least one primary SIC code and, in some cases, one or more overflow SIC codes. All relevant descriptions appear in the primary column, with a few additional ones found only in the overflow columns.

This means some ABI and address entries may be associated with more than one religious category. Additionally, ABIs with multiple addresses may file under different religious codes, either because they are interfaith organizations or because the codes are being used more generally. We therefore assessed classification consistency across entries; specifically, whether any entries are filed under multiple religious categories and how frequently this occurs.

Consistencies by classifiers defined in "Results/From Clean Raw Data/Step 3_2026 Format/SIC Code Classifications_08.15.2026.csv" were matched across all primary and overflow SIC columns separately. Any SIC classifiers associated with an ABI matching this criteria were then reported to show how many additional religious or interfaith classifications were present. This was done using the custom `function sic_desc_split_summary_dt()`.

Classifiers were assigned based on clear attribution to a specific religion. However, a few descriptions showed a surprising degree of overlap across religions, the most notable example being the SIC description "CHURCHES". Others, such as "RELIGIOUS ORGANIZATIONS", "PLACES OF WORSHIP", and "CONVENTS & MONASTERIES", were too vague to attribute to a single religion and were therefore labeled "Interfaith" to reflect their use across different religious contexts.

Classifiers are assigned based on the primary and overflow SIC codes. If, at the ABI level, only one religious classifier appears alongside "Interfaith" entries, all entries for that ABI are assigned that religion. This override is only applied when a single religion is identified; ABIs with multiple religions listed retain their original classifications.

Some "Interfaith" descriptions, such as "CHURCHES" and "CHURCH ORGANIZATIONS", are primarily associated with Christian organizations and are therefore disambiguated to "Christian Church" in the absence of other clear religious identifiers. Otherwise, they are treated the same way as the other interfaith categories. This approach may miss cases where these descriptions

apply to interfaith ministries, but will reduce the overattribution of interfaith classifications, which prior results suggest is likely to occur. An exception is made for entries with the SIC description “SYNAGOGUES MESSIANIC”, which are attributed to both “Christian Church” and “Jewish Synagogue”.

Following this, entries that could not be disambiguated alone were labeled “Interfaith”, as they may be resolvable in the presence of other, clearer classifiers. It is important to clarify, however, that these remaining “Interfaith”-only cases are actually unspecified. An “Interfaith” column was also created to represent all cases where more than one religion, including “Other” and “Unspecified”, is TRUE for a given entry.

Fill Minor Years Reported Gaps

A closure is defined as an event in which four or more consecutive years have no filings under any address. To simplify metric calculations, these missing years will be filled in. Care must be taken to ensure that this process does not induce duplicate records in the event of temporary relocations within intervening years.

In the first pass, the custom function `fill_gaps_leq_k()` fills all qualifying gaps identified in a row-wise procedure. Some of these fills coincided with a temporary relocation entry, causing column-wise sums over the binary year columns to no longer be unique (> 1). These entries are corrected using the custom function `fix_bad_abi_flagfill()`, which performs a row-wise gap-fill for each impacted ABI using a sentinel flag value of 50. Any flagged fills that fall in an ABI-year where other non-zero values already exist are then pruned to avoid introducing conflicts. Finally, remaining flagged values are converted to 1, confirming the gap as a legitimate fill.

Identifying and Quantifying Moves vs. Reopenings at New Location

Moves are defined as any change of address, including returns to a previous address. However, this excludes cases where more than four years elapsed between point a and point b. In such cases, the event is considered a reopening at a new location rather than a move.

Address changes are assessed between each pair of sequential addresses recorded for an ABI. The distance in kilometers between the current and previous address is computed using the Haversine formula via `geosphere` [26], which calculates the shortest surface distance between two geographic coordinates. If the intervening years between sequential addresses exceed

four, the transition is relabeled as a “Reopened (New Location)”. Additional indicator columns are generated as boolean flags identifying whether the move exceeded 5, 10, or 25 miles.

Some addresses were moved from, returned to, and moved from again multiple times, causing the results to no longer correspond one-to-one with the ABI–address unique entry design. These entries were therefore summarized to the address level, with move counts aggregated, distances summarized as mean and maximum, boolean flags resolved as any-true, and string fields (e.g. moved to addresses) collapsed into semicolon-separated values.

Additional details regarding the algorithm’s performance outcomes can be found in Results - Step 3: Standardize and Reshape Metadata to Wide Format, which includes results specifically assessing the outcomes of the three quality control assessments.

Results

Step 1: Nomenclature Standardization and Parquet Conversion

Processing the NAICS code columns was straightforward, involving the creation of new columns from the trailing two-digit proprietary codes and the renaming of the six-digit NAICS column names.

Testing the SIC code assumptions revealed points where standardization was needed. Overall, 18 description pairs with minor typographical variations were reconciled to a single version, 2 code redundancies were reduced, and 4 sources of `NA` descriptions were reconciled to existing descriptions for the same code.

Refer to Appendix - Primary SIC Codes and Descriptions All SIC Codes and Descriptions, and Step 1 Final Result for information on where to find the code that generates these results, as well as the results codebook.

Step 2: Clean and Validate Addresses and Geolocation, and Annotate Records with Census Boundaries

Compilation Quality Control Checks

Refer to Appendix - Step 2 Final Result - Import Quality Checks, State-Level Shapefile for Basemap, and Missing Address Lines and PO Boxes By City for information on where to find the code that generates these results, as well as the results codebook.

Results were compiled both from arrays within a batch and between batches. To ensure there was no data loss or erroneous introduction of duplications, basic importation QCs were generated. **SUBSECTION D2: Confirm Complete ABI Coverage** confirms that all imported data contains the needed ABIs — those that filed within the US. **SUBSECTION D4: At-Point-of-Evaluation Quality Checks** for the Address QC dataset further confirms that address validation and matching did not create any duplications of records.

Figure 3 and Figure 4 below summarize discrete outcomes for the different cleaning and validation steps conducted by array. Unlike in the respective QC datasets discussed in subsequent sections, these values are pulled from the main dataset where all ABIs are represented.

Figure 4 shows most entries resulted in a GEOID match (block group, tract, county, and state) averaging over 95%. On average, 80% of the geocoordinates used for spatial joining were verified, with less than 10% not comparable with the database due to missing `address_line_1`. Notably, the second batch array, where no address verification was allowed, yielded some arrays that had close to 100% unverified geocoordinates.

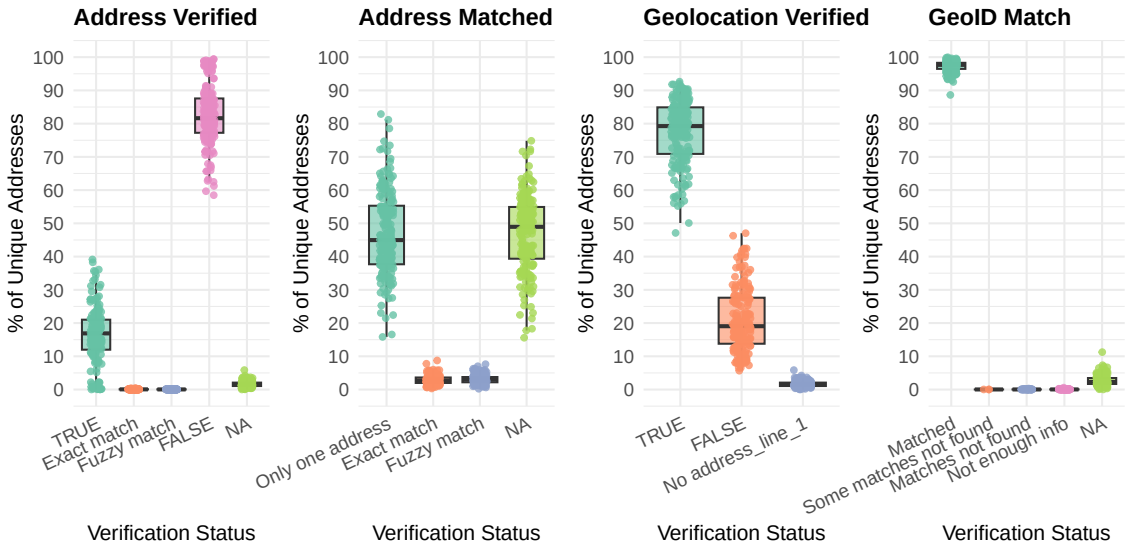
> **Note**
>
> NAs for both verified and matched addresses were further resolved to those caused by addresses missing a valid `address_line_1` after importing. Results are not shown in these import QC tables.

Less than 20% of addresses were verified, likely attributable to the API query limitations imposed. Less than 5% of addresses were matched either exactly or fuzzy matched, with most matching occurring in the agnostic algorithm and negligible matching resulting from unverified addresses reconciled to verified ones.

Figure 3 resolves how many entries did not yield a match to any type of census boundary by decennial year. For some of these columns, such as the CBSA and CSA columns, missingness

QC Flag Distributions Across Arrays

Batch 18850425: 87% ABI coverage (allowed address verification)



QC Flag Distributions Across Arrays

Batch 20823868: Remaining ABI not covered in batch 18850425 (no address verification)

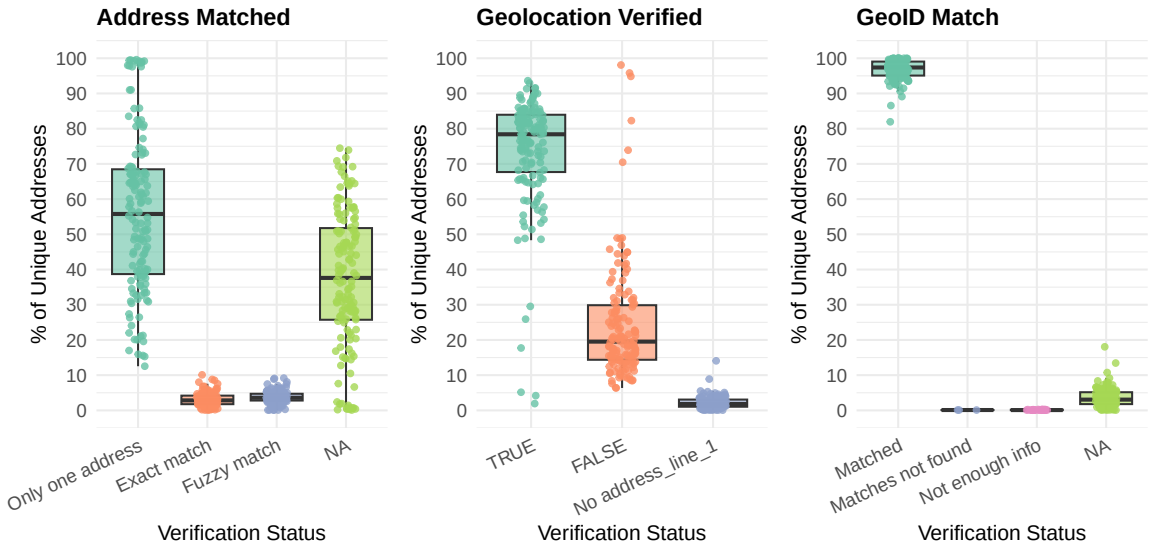
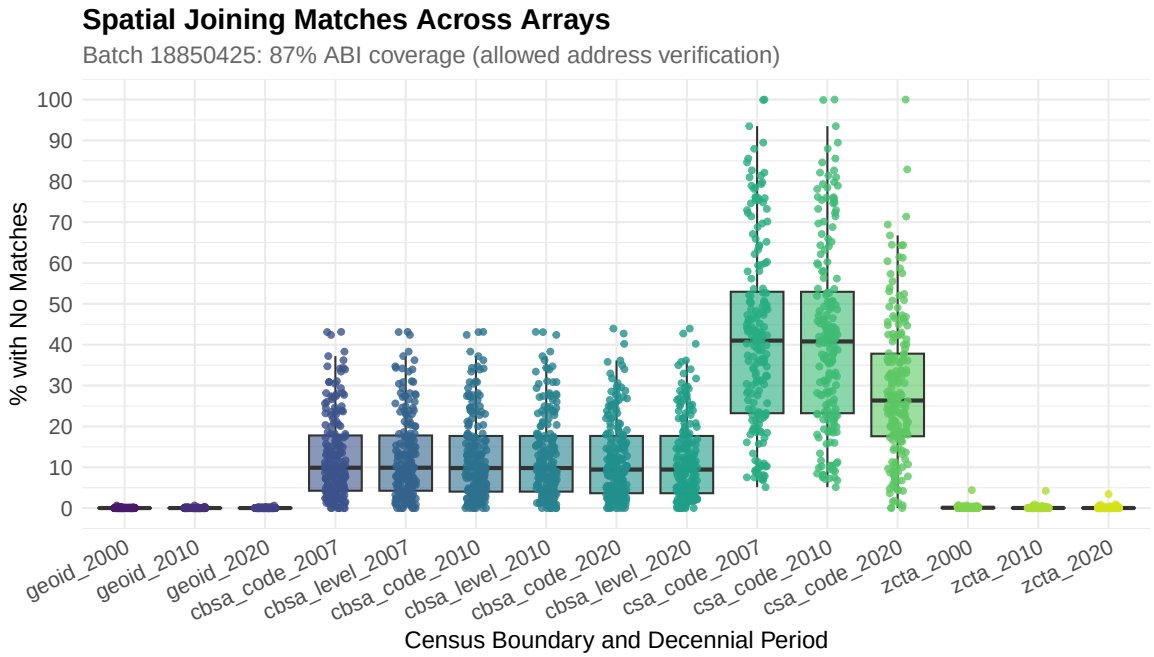


Figure 3


Spatial Joining Matches Across Arrays
Batch 18850425: 87% ABI coverage (allowed address verification)
% with No Matches
100
90
80
70
60
50
40
30
20
10
0
geoid_2000
geoid_2010
geoid_2020
cbsa_code_2007
cbsa_level_2007
cbsa_code_2010
cbsa_level_2010
cbsa_code_2020
cbsa_level_2020
csa_code_2007
csa_code_2010
csa_code_2020
zcta_2000
zcta_2010
zcta_2020
Census Boundary and Decennial Period

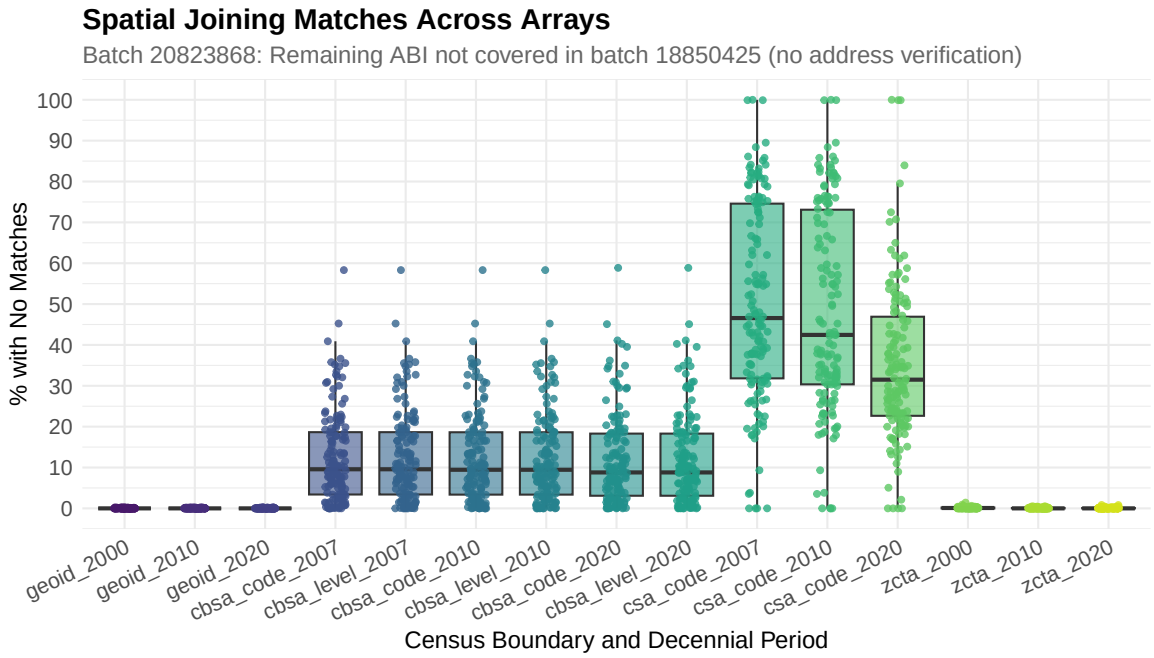

Spatial Joining Matches Across Arrays
Batch 20823868: Remaining ABI not covered in batch 18850425 (no address verification)
% with No Matches
100
90
80
70
60
50
40
30
20
10
0
geoid_2000
geoid_2010
geoid_2020
cbsa_code_2007
cbsa_level_2007
cbsa_code_2010
cbsa_level_2010
cbsa_code_2020
cbsa_level_2020
csa_code_2007
csa_code_2010
csa_code_2020
zcta_2000
zcta_2010
zcta_2020
Census Boundary and Decennial Period

Figure 4

is expected since these census boundaries are assigned to urban locations. Both batches show very high annotation rates (low missingness) across the GEOID columns, with some limited missingness in the ZCTA columns. CBSA code and level missingness remains similar on average across the decennial years, but CSA code missingness attenuated from the 2000 to 2010 and again to the 2020 decennial period.

Overview of Address Completeness and ABI Filing Patterns

The cleaning and validation conducted in **Step 2** relied on a few core assumptions. Where possible, missing data or inconsistencies were handled by the algorithm, limiting the number of entries removed at this stage and maximizing the number of records rendered usable for downstream analysis. Special Considerations for PO Boxes and Missing `address_line_1` Entries examined the biggest factors to consider and outlined when critically missing values justify removing an entry from downstream analysis.

Table 3: Distribution of Number of Unique Addresses ABI Filed (counts reported in thousands)

	1	2	3	4	5	6	7	8	9	10	11	12
Count (K)	759	205	74	28	10	3	0.900	0.300	0.050	0.010	0.002	0.002
Percent	70.21	19.00	6.88	2.59	0.91	0.29	0.09	0.02	0.00	0.00	0.00	0.00

In addition, a few other variable characteristics were assessed and assumptions about them tested. All years reported are unique within a given ABI, making all ABI and address combinations unique for a given years-open value. 70% of ABIs filed under one unique address for the entire span of time, with 10% moving more than twice and less than 1% moving more than five times (Table 3).

City, state, five-digit ZIP code, and four-digit ZIP code extension are also given with an `address_line_1`. No entries are missing state or city; 0.001% lack a five-digit ZIP code and $\approx$ 22% lack the four-digit ZIP extension. These missing values pose minimal geocoding risk, except for the 8 entries without a ZIP code, for which address validation failure would preclude SimpleMaps secondary validation. All entries have consistent formatting, assessed by string character length, for the state two-digit acronym, five-digit ZIP code, and four-digit extension, when reported.

Address Validation and Consolidation Process

Refer to Appendix - Step 2 Algorithm Quality Check Datasets - Import Quality Checks and Step 2 Algorithm Quality Check Datasets - Addresses for information on where to find the code that generates these results, as well as the results codebook.

The most comprehensive quality check (QC1) contains all ABIs; the remaining checks may show missing ABIs due to records being skipped during validation or matching (Table 4). Common reasons include `NA` values for `address_line_1` or no addresses remaining after USPS API validation. Additionally, approximately 70% of ABIs are expected to be ineligible for agnostic matching; without prior validation, these ABIs will not be captured by this metric. Proportions of ABIs that did not qualify for QC2 or QC3 were consistent across batches.

Table 4: Missingness test outcomes by QC key. **TRUE/FALSE** are counts of arrays passing/failing; **Min/Mean/Max** are the min/avg/max % of ABIs missing in those arrays.

`import_qc_18`

Key	FALSE	TRUE	Min	Mean	Max
qc1	0	188	0.00	0.00	0.00
qc2	187	1	0.00	1.64	11.60
qc3	187	1	0.00	1.64	11.60

`import_qc_20`

Key	FALSE	TRUE	Min	Mean	Max
qc1	0	142	0.00	0.00	0.00
qc3	119	23	0.00	1.76	14.4

QC1: API Usage, Matching Overview, and Unique Result Consistency

Of the records that underwent API validation, approximately 1.5% did not attempt verification — `verified_address` is absent from the dataset for these records (Table 5). All such cases correspond to a `NA` value in `address_line_1`; direct examination of these ABIs confirmed that no valid address was available, causing the validation step to be skipped.

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Table 5: Verification Attempt by NA Address Presence and API Permission (%)

Allow Verification: `FALSE`

	Attempted	
Any NA Addresses	**FALSE**	**TRUE**
FALSE	12.66	0.00
TRUE	0.38	0.00

Allow Verification: `TRUE`

	Attempted	
Any NA Addresses	**FALSE**	**TRUE**
FALSE	0.00	84.71
TRUE	1.43	0.82

Almost all ABIs (~98%) were processed through agnostic matching. No duplicates were detected following this step, and no address-year combinations differed between the raw and validated forms of the data, confirming that no data loss occurred.

QC2: Address Validation Using the USPS API

As described in LOOP PART B: Consolidate and Verify the Addresses, unverified addresses were compared against verified ones using either exact or fuzzy matching. Exact matches on `address_line_1` bypassed the geolocation difference test entirely, while fuzzy matches required both coordinate differences to fall below 0.02 degrees (approximately 2 kilometers, or $\approx$ 1.4 miles). Of addresses that failed verification, fewer than 1% underwent fuzzy matching and subsequently failed the geolocation test (Table 6).

Table 6: Verified Address by Geolocation Test (% column)

Geolocation Test	Verified Address				
	TRUE	Exact Match	Fuzzy Match	FALSE	No `address_line_1`
TRUE	0.00	0.00	100.00	0.00	0.00
FALSE	0.00	0.00	0.00	0.11	0.00
Override	0.00	100.00	0.00	0.00	0.00
`NA`	100.00	0.00	0.00	99.89	100.00

Among addresses that were matched exactly or by fuzzy matching, most failed USPS verification due to bad or access denied requests. Rate limiting accounted for fewer than one in five of these verification failures. In contrast, addresses that failed verification and could not be matched to any verified address were predominantly blocked by rate limits (Table 7).

Table 7: Verified Address by USPS Status Detail (% column)

USPS Status Detail	Verified Address				
	TRUE	Exact Match	Fuzzy Match	FALSE	No `address_line_1`
200 Successful operation	100.00	0.00	0.00	0.00	0.00
400 Bad request	0.00	77.56	82.87	1.54	0.00
403 Access denied	0.00	1.55	1.77	23.07	0.00
429 Rate limit reached	0.00	20.89	15.35	75.38	0.00
`NA`	0.00	0.00	0.00	0.00	100.00

The vast majority of successful verification attempts happened on the first try, with less than

1% verified using the secondary fallback strategy (Table 8). NOTE: All NAs correspond to outcomes where `attempt_succeeded == "No address_line_1"`.

Table 8: Attempt Succeeded by USPS Status Detail (% row)

	Attempt Succeeded		
USPS Status Detail	**First Try**	**With City Correction by Zip5**	**None**
200 Successful operation	99.92	0.08	0.00
400 Bad request	0.00	0.00	100.00
403 Access denied	0.00	0.00	100.00
429 Rate limit reached	0.00	0.00	100.00

QC3: Address Matching Results

Most fuzzy-matched attempts (86%) passed the geolocation test; fewer than 1% passed with an incomplete comparison due to one record missing geocoordinates (Table 9). This implies a small number of entries with missing geocoordinates were supplemented through string matching, in addition to any resolved through address-based geocoding.

Table 9: Geolocation Test Outcome by Matched Address (% row)

	Geolocation Test			
Matched Address	**PASS**	**PASS; no Lon/Lat**	**FAIL**	**Override**
Exact match	0.00	0.00	0.00	100.00
Fuzzy match	86.28	0.02	13.70	0.00
Only one address	0.00	0.00	0.00	0.00

Address-Based Geocoding

Refer to Appendix - Step 2 Algorithm Quality Check Datasets - Import Quality Checks and Step 2 Algorithm Quality Check Datasets - Geocoordinates for information on where to find the code that generates these results, as well as the results codebook.

The vast majority of arrays contained ABIs that were not processed through the geolocation tests (Table 10). This is expected, since QC1 and QC2 restricted their assessments to only those ABI records where an address was available for address-based geocoding. QC2 additionally

filtered by non-missing verified geocoordinates. QC3 focused only on ABI records where more than one valid pair of geocoordinates was available for generating the summary tables.

Lack of representation is even higher for the second batch run, with maxima as high as 98% for QC2. However, the averages excluded still tend to be comparable with the previous batch. This is consistent with the results seen in earlier plots, where it was shown that this batch resulted in some arrays with high degrees of no validation (Figure 3 and Figure 4).

Table 10: Missingness test outcomes by QC key. **TRUE**/**FALSE** are counts of arrays passing/failing; **Min**/**Mean**/**Max** are the min/avg/max % of ABIs missing in those arrays.

	import_qc_18						import_qc_20				
Key	**FALSE**	**TRUE**	**Min**	**Mean**	**Max**	**Key**	**FALSE**	**TRUE**	**Min**	**Mean**	**Max**
qc1	187	1	0.00	1.64	11.60	qc1	119	23	0.00	1.76	14.40
qc2	188	0	2.40	13.80	44.60	qc2	142	0	0.20	16.90	98.10
qc3	188	0	1.44	15.40	32.90	qc3	141	1	0.00	13.30	36.40

QC1: US Census Bureau Geocoder API Validation

A higher proportion of addresses with valid geocoordinates (non-NA) were verified through address-based geocoding. ≈ 80% of entries without valid geocoordinates failed to verify in comparison to ≈ 22% of those with valid geocoordinates (Table 11).

Table 11: Enough Valid Geo by Geo Verified (% column)

	Enough Valid Geo	
Geo Verified	**FALSE**	**TRUE**
TRUE	10.30	78.12
FALSE	79.13	21.18
No address_line_1	10.57	0.70

"N Addresses" counts the number of unique addresses were aggregated and had valid geocoordinates (Table 12). In general, we see an increase in success as more records are included, but this drops down again with more than 7 records, with 8 having a 50/50 match rate.

As would be expected, any query where the number of attempts falls below the max number of benchmark/vintage tries listed successfully verified geocoordinates. ≈ 98% of queries were successful with the first try. The next query tries with some successes was the third set, with

Table 12: Geo Verified by Number of Addresses (% column)

	N Address								
Geo Verified	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
TRUE	10.30	76.04	85.52	87.58	87.00	88.21	85.86	70.83	50.00
FALSE	79.13	23.08	14.48	12.42	13.00	11.79	14.14	29.17	50.00
No address_line_1	10.57	0.88	0.00	0.00	0.00	0.00	0.00	0.00	0.00

only 0.02% verifying using the fourth set (Table 19). Recall the benchmark and vintage pairing tries listed under **SUBSECTION A3: Set Geocoder Search Priorities**:

Table 13: Benchmark and Vintage Try Pairs

benchmark_name	vintage_name
Public_AR_Census2020	Census2020_Census2020
Public_AR_Census2020	Census2010_Census2020
Public_AR_Current	Current_Current
Public_AR_ACS2025	Current_ACS2025

The first in queue, `[Public_AR|Census2020]_Census2020`, is expected to reference the same database as the third, `[Public_AR|Current]_Current`, supporting the higher matches observed using both.

Table 14: Geolocation verification status by try attempt (% column)

	Geo Verified		
# Attempts	**TRUE**	**FALSE**	**No address_line_1**
1	97.70	0.00	0.00
2	0.10	0.00	0.00
3	2.18	0.00	0.00
4	0.02	100.00	0.00
<NA>	0.00	0.00	100.00

Of outcomes that reached four tries, 45 different combinations of API query responses were received, with 99.5% having all 200 status codes, indicating a successful query interaction. This confirms that failures to match largely were a result of no match being available in the respective benchmark/vintage try pairings databases used.

Table 16: Address verified and address matched outcomes by geolocation verified (% row)

	Address Verified			Address Matched		
Geocoordinates Verified	**FALSE**	**Other**	**TRUE**	**FALSE**	**Other**	**TRUE**
FALSE	75.10	15.60	9.33	60.90	31.20	7.90
No address_line_1	0.00	100.00	0.00	0.00	100.00	0.00
TRUE	74.00	12.80	13.10	41.60	46.20	12.20

A string comparison was done between the queried address and the matched address in the US Census Bureau database. The algorithm processing JSON responses handled multiple possible matches through best match triaging. This assessment quickly identifies if the given address and best match chosen from the JSON response are exactly the same or similar (δ = 0.15). Results show most JSON results were the exact same, 78% (Table 15). Almost 19% of responses that were not exactly the same were similar. Only 3% were neither similar nor the exact same.

Table 15: Query and JSON match address comparison (% column)

	Address Same	
Address Similar	**FALSE**	**TRUE**
FALSE	3.28	0.00
TRUE	18.70	78.02

It is important to assess if verifying addresses or agnostically matching them results in any changes with address-based geocoding. Results where a verification was retained (directly or through matching) show no obvious difference between those that had geocoordinates verified and those that did not (Table 16).

For both verified addresses and agnostic matching, those compressed via fuzzy matching saw an attenuation in retrieving a match (Table 17). Exact matching did not seem to change results, neither compared with those directly verified by address nor with results having only one address (no matches attempted).

Table 17: Geo Verified by Address Verified/Matched (% row).

Address Verified	TRUE	FALSE
TRUE	83.86	16.14
Exact match	86.91	13.09
Fuzzy match	65.43	34.57
FALSE	78.41	21.59
<NA>	74.63	24.57

Address Matched	TRUE	FALSE
Exact match	88.15	11.85
Fuzzy match	80.78	19.22
Only one address	84.51	15.49
<NA>	71.56	28.44

QC2: Reported vs. Validated Geocoordinate Comparison

It would be interesting to assess how well the reported geocoordinates aligned to the verified match from the Census Bureau Geocoder database. It was observed that 21% had a difference between at least one geocoordinate of more than 0.002 degrees (about 2 city blocks). 12% had a difference between both.

Table 18: Latitude/Longitude difference relative to 0.002 degrees (%)

	Lon Diff > 0.002		
Lat Diff > 0.002	FALSE	TRUE	<NA>
FALSE	78.72	5.49	0.00
TRUE	3.49	12.30	0.00
<NA>	0.00	0.00	0.00

When looking only at those where the differences exceeded 0.002 degrees, the mean difference is 0.0197 degrees and the max is 22.54 degrees in latitude; the mean difference is 0.0233 degrees and the max is 75.72 degrees in longitude. While most of the records were close to 2 kilometers or 1.4 miles difference, some were as many as 22 degrees, about the distance from New Haven, CT to Mexico City, Mexico, or even 75 degrees, about the distance from New Haven to London, England.

QC3: Geocoordinate Variation by Address

It would be interesting to assess how well the reported values aligned within the same reported address. Based on limited manual assessment, it was noted that these varied. Most addresses were within 0.02 degrees longitude or latitude, about 2 kilometers or 1.4 miles. About 5%, however, had some variation, either in longitude, latitude, or both (Table 19).

Table 19: Latitude/Longitude spread relative to 0.02 degrees (%).

	Lon Spread > 0.02	
Lat Spread > 0.02	**FALSE**	**TRUE**
FALSE	95.37	1.69
TRUE	1.02	1.92

One concern is that address validation or matching increased the amount of geolocation variation. However, based on the results from the previous quality check, this may be moot since the given coordinates have a large margin of error.

Looking at the latitude for verified addresses, there does not appear to be any strong effect on the max or average difference of geolocation amongst the same address. Similarly, for agnostically matched addresses, there does not appear to be a strong discernible influence on the max or average difference of geolocation amongst the same address (Table 20).

Looking again but for longitude, there also does not appear to be a strong influence caused by verifying the address or agnostically matching them (Table 20). For context, the number of results represented for each stratum is included. These run a little on the high end (Table 12).

Table 20: Lat/Lon spread summary (> 0.02) (subset where spread flag is TRUE)

Panel A: By Any Address Verified

	Lat Spread > 0.02				Lon Spread > 0.02			
Any Verified	**n**	**Max Diff**	**Mean Diff**	**Avg N Geo**	**n**	**Max Diff**	**Mean Diff**	**Avg N Geo**
FALSE	23433	51.8	0.0538	13	28926	110.0	0.0659	13
TRUE	3972	47.8	0.0678	14	4698	96.0	0.0835	13
<NA>	5660	47.4	0.0502	12	6997	102.0	0.0526	11

Panel B: By Any Address Matched

	Lat Spread > 0.02				Lon Spread > 0.02			
Any Matched	**n**	**Max Diff**	**Mean Diff**	**Avg N Geo**	**n**	**Max Diff**	**Mean Diff**	**Avg N Geo**
FALSE	839	46.0	0.0646	7	939	98.4	0.0813	7
TRUE	10741	45.7	0.0497	17	13256	99.6	0.0634	17
<NA>	21485	52.2	0.0570	11	26426	110.0	0.0662	11

Version: 006.07.15.26
Generated using Quarto (v. 1.8.27)
and RStudio (v. 2026.01.1+403)

Census Boundary Annotation Using Point-in-Polygon Spatial Joins

Refer to Appendix - Step 2 Algorithm Quality Check Datasets - Import Quality Checks and Step 2 Algorithm Quality Check Datasets - Census for information on where to find the code that generates these results, as well as the results codebook.

The vast majority of arrays contained ABIs that were not processed through the QC1 census tests assessing GEOIDs, where a higher proportion of arrays completely represented ABIs when assessing the CBSA/CSA/ZCTA boundaries (Table 21). This is expected, since QC1 restricted its assessments to only ABI records where enough geocoordinates were available for spatial joining. QC1 also filters by entries where a GEOID match was made after joining.

Lack of representation is slightly higher for the second batch run, with maxima as high as 19% for QC1 as opposed to 16%. However, the averages excluded still tend to be comparable with the previous batch. This is consistent with the results seen in earlier plots, where it was shown that this batch resulted in some arrays with high degrees of `NA` after spatial joining (Figure 3 and Figure 4).

Table 21: Missingness test outcomes by QC key. **TRUE**/**FALSE** are counts of arrays passing/failing; **Min**/**Mean**/**Max** are the min/avg/max % of ABIs missing in those arrays.

`import_qc_18`

Key	FALSE	TRUE	Min	Mean	Max
qc1	188	0	0.02	2.61	16.30
qc2	77	111	0.00	0.03	0.64

`import_qc_20`

Key	FALSE	TRUE	Min	Mean	Max
qc1	126	16	0.00	2.91	19.0
qc2	26	116	0.00	0.03	0.30

QC1/2: GEOID and Core Statistics Vintage Alignment

A quick check comparing dimensions of distinct outcomes over address and over address with census boundaries showed each unique address is not consistently associated with the same census metadata for either QC1 or QC2. This result could be skewed due to the matching processes done. In general, we have reason to believe the same census boundaries were not applied to the same address over the years.

RStudio Console

```
## SUBSECTION D4: At-Point-of-Evaluation Quality Checks"

# GEOID columns ~ line 2909
n_distinct(census_qc$qc1$address) == nrow(distinct(census_qc$qc1, address,
↪  census_block))
n_distinct(census_qc$qc1$address) == nrow(distinct(census_qc$qc1, address,
↪  census_tract))
n_distinct(census_qc$qc1$address) == nrow(distinct(census_qc$qc1, address,
↪  county_code))
n_distinct(census_qc$qc1$address) == nrow(distinct(census_qc$qc1, address,
↪  fips_code))

# Statistical area columns ~ line 2944
n_distinct(census_qc$qc1$address) == nrow(distinct(census_qc$qc1, address,
↪  cbsa_level))
n_distinct(census_qc$qc1$address) == nrow(distinct(census_qc$qc1, address,
↪  cbsa_code))
n_distinct(census_qc$qc1$address) == nrow(distinct(census_qc$qc1, address,
↪  csa_code))
```

In this section, the big questions we want to answer are these: was the raw data annotated with the correct census boundaries, and if not, what is the extent of the error; do the matched boundaries by decennial year correspond to the years the address was filed under. Note that the data provider did not include ZCTA columns, which is why these alignments are not checked here.

An attenuation of matching is observed as the boundary becomes smaller, with near perfect matching for FIPS and county. Only 0.01% of tract and block entries were reported as `NA`, and are consequentially uncheckable. CBSA code yielded a high match rate at 85%, CSA at 65%, and CBSA level at the lowest, with matches at 3%. Many CBSA level results were uncheckable too, at 12% (Table 22).

Table 22: Do raw values match any spatially joined decennial year equivalent? (% column)

`census_qc$qc1`

	Matched	None	Uncheckable
FIPS	98.56	1.44	0.00
County	97.63	2.37	0.00
Tract	84.64	15.35	0.01
Block	84.16	15.82	0.01

`census_qc$qc2`

	Matched	None	Uncheckable
CBSA Code	85.87	14.13	0.00
CBSA Level	3.14	84.50	12.36
CSA	65.35	34.65	0.00

As would be expected, FIPS and county generally matched with GEOIDs across decennial periods (over 97%). Tract and block saw 44% and 35%, respectively. The next big category of matching was the 2000 decennial period, but nearly the same proportion of records match over the 2000, 2010 and 2010, 2020 decennial periods (Table 23). This indicates that the given GEOID boundaries do not clearly follow a pattern of annotating information from any one decennial period.

Most CBSA code, CBSA level, and CSA records did not match a decennial year, with a high level reporting as `NA`. The `NA` values arise from entries failing to annotate with that boundary during spatial joining, which is expected since these represent urban areas and will not cover the whole contiguous US. The vintage with the most matches was the 2020 vintage (as high as 14%), with all other vintages ranking well below 1% (Table 23).

Table 23: Records whose matched vintage set equals the given row label (% rows)

`census_qc$qc1`

Vintages	FIPS	County	Tract	Block
2000	0.00	0.01	12.71	14.85
2010	0.00	0.00	2.48	4.22
2020	0.02	0.02	5.01	6.09
2000, 2010	0.02	0.03	10.44	10.67
2000, 2020	0.00	0.00	0.02	2.31
2010, 2020	0.03	0.02	9.56	10.61
2000, 2010, 2020	98.48	97.54	44.42	35.42
None	1.44	2.37	15.35	15.82
Not reported	0.00	0.00	0.00	0.00
Uncheckable	0.00	0.00	0.01	0.01

`census_qc$qc2`

Vintages	CBSA Code	CBSA Level	CSA
2000	0.00	0.00	0.00
2010	0.00	0.00	0.02
2020	5.40	0.47	13.53
2000, 2010	0.00	0.00	0.00
2000, 2020	0.00	0.00	0.00
2010, 2020	0.33	0.01	0.01
2000, 2010, 2020	0.00	0.00	0.00
None	14.13	84.50	34.65
Not reported	0.00	0.00	0.00
Uncheckable	0.00	12.36	0.00
<NA>	80.13	2.66	51.79

The variability in census boundaries indicated earlier may legitimately be a manifestation of census boundaries getting correctly applied when that address was filed. However, we see that this assumption only holds for the most macro-level boundaries, FIPS and county. High adherence is also observed for CBSA codes, despite most matching with the 2020 decennial year (Table 24).

Granular boundaries like tract and block showed progressively increasing lack of alignment, with as high as 15% being unable to assess (Table 24). CSA had just more than half align, but users should note that this proportion may be misleading. We would expect that there is a high degree of failed comparisons since CSA is not applied for all parts of the US. Therefore, the real proportion of candidates is expected to be higher.

Table 24: Vintage-alignment outcomes with years open (% row)

`census_qc$qc1`

Boundary	FALSE	TRUE	<NA>
FIPS	0.04	98.52	1.44
County	0.02	97.60	2.37
Tract	17.55	67.09	15.36
Block	23.26	60.90	15.84

`census_qc$qc2`

Key	FALSE	TRUE	<NA>
CBSA Code	5.07	80.80	14.13
CBSA Level	1.22	1.92	96.86
CSA	12.11	53.24	34.65

Step 3: Standardize and Reshape Metadata to Wide Format

Normalize SIC Codes and Annotate with Classifiers

4% of unique ABI/address entries have more than one primary SIC code associated. Notably, only 14% of these cases include a verified or matched address, indicating that the discrepancy predates the redundancy reduction steps in this pipeline. This raises the question of whether the variation reflects changes in religious or denominational affiliation over time. This question falls outside the scope of the current assessment and will not be explored further, only noted here.

On average, entries with multiple primary SIC codes had only a second one, with some having as many as five unique primary codes. Most entries with overflow SIC codes had only one, with some having as many as eleven.

Assign Religious Classifiers

SIC code classifiers "Results/From Clean Raw Data/Step 3_2026 Format/SIC Code Classifications_08.15.2026.csv" were defined using descriptions that clearly or implicitly indicate the religion a business is associated with. The SIC descriptions "SYNAGOGUES MESSIANIC" and "TEMPLES BUDDHIST EDUCATIONAL INSTITUTION" are the only classifiers absent from the primary SIC code column; all others appear in both primary and overflow SIC columns.

Table 25 shows a selection of results assessing classification consistency by religious classifier across both primary and overflow columns. These results highlight the top descriptions in entries associated with a given religion over the primary SIC columns. Some classifiers that

are not clearly attributable to a single religion show a high degree of overlap. The most notable example is the SIC description “CHURCHES”.

Full results can be found in “Results/From Clean Raw Data/Step 3_2026 Format/”.

Table 25: Top primary SIC-description outcomes (% of search hits)

Christian Church

x	classifier	Freq
CHURCHES	Christian Church; Interfaith	46.4
RELIGIOUS ORGANIZATIONS	Interfaith	21.5
CHURCH ORGANIZATIONS	Christian Church; Interfaith	10.7
CLERGY	Interfaith	8.9
MINISTRIES-OUT REACH	Interfaith	3.4

Muslim Mosque

x	classifier	Freq
MOSQUES	Muslim Mosque	44.3
CHURCHES	Christian Church; Interfaith	32.3
RELIGIOUS ORGANIZATIONS	Interfaith	15.8
MOSQUES MUSLIM-ISLAMIC CHARITABLE-ORGZTN	Muslim Mosque	5.1
CHURCH ORGANIZATIONS	Christian Church; Interfaith	1.3

Jewish Synagogue

x	classifier	Freq
SYNAGOGUES	Jewish Synagogue	46.4
SYNAGOGUES JEWISH	Jewish Synagogue	31.2
CHURCHES	Christian Church; Interfaith	7.1
RELIGIOUS ORGANIZATIONS	Interfaith	4.3
SYNAGOGUES ORTHODOX	Jewish Synagogue	3.6

Hindu Mandir

x	classifier	Freq
CHURCHES	Christian Church; Interfaith	43.5
TEMPLES-HINDU	Hindu Mandir	43.5
SYNAGOGUES	Jewish Synagogue	8.7
RELIGIOUS ORGANIZATIONS	Interfaith	2.2
TEMPLES-BUDDHIST	Buddhist Temple	2.2

Most entries were associated with a Christian church (76%). The next largest category was Unspecified at approximately 22%, followed by Judaism at 1.7%. All other religions fell below 1% (Table 26).

Table 26: Religion classification flags by ABI (percent of ABIs).

Religion	FALSE (%)	TRUE (%)
Buddhist Temple	99.89	0.11
Christian Church	23.60	76.40
Hindu Mandir	99.97	0.03
Jewish Synagogue	98.31	1.69
Muslim Mosque	99.88	0.12
Sikh Gurdwara	99.99	0.01
Other Religion	99.62	0.38
Interfaith	99.93	0.07
Unspecified	78.39	21.61

Fill Minor Years Reported Gaps

9.2% of ABI/address entries required gap filling, and approximately 3% of ABIs had an induced duplicate record as a result. For most ABIs, only one qualifying gap was filled (89%), of which 62% were 1-year gaps, 25% were 2-year gaps, and 8% were 3-year gaps. Table 27 shows that as the number of gaps filled increases, so does the average gap length. Note that zeros are omitted for readability and do not indicate missing values.

Table 27: Gaps filled by average filled-gap length (row %).

	Gaps Filled					
Avg Length Filled	1	2	3	4	5	6
1	65.05	42.52	25.29	19.51	20.00	
1.2					20.00	
1.25				30.89		
1.33			33.71			100.00
1.4					20.00	
1.5		35.23		28.46		
1.6					40.00	
1.67			22.39			
1.75				11.38		
2	25.79	16.81	12.64	8.13		
2.25				1.63		
2.33			4.65			
2.5		4.53				
2.67			1.23			
3	9.16	0.92	0.09			

Version: 006.07.15.26
Generated using Quarto (v. 1.8.27)
and RStudio (v. 2026.01.1+403)

Identifying and Quantifying Moves vs. Reopenings at New Location

Approximately 26% of entries involved a move or reopening at a new location, of these about 40% involved a PO Box. Most moves were less than 5 miles, with only approximately 14% exceeding 5 miles (Table 28). Most relocations were also less than 5 miles, with approximately 1% exceeding 5 miles. As expected, the share of addresses beyond 10 and 25 miles decreased incrementally: approximately 5% and 1%, respectively. Note that "reopenings" here refer to those that reopened at a new location following a closure.

Table 28: Move distance indicators by transition type (% row)

	> 5 mi	
	FALSE	TRUE
Moved	86.18	13.82
Reopened	82.74	17.26

	> 10 mi	
	FALSE	TRUE
Moved	94.51	4.47
Reopened	0.95	0.07

	> 25 mi	
	FALSE	TRUE
Moved	97.84	1.15
Reopened	1.00	0.02

Version: 006.07.15.26
Generated using Quarto (v. 1.8.27)
and RStudio (v. 2026.01.1+403)

Appendix

All SIC Codes and Descriptions

For the implementation, refer to **PART B: Correct the SIC Nomenclature** in "Code/Clean Raw Data_Step 1_2026 Format.R". The data processed was the raw export `church_long_form_050926.csv`. Results were written to `"./Data/Results/KEEP LOCAL/From Process Data Update/Non-Primary SIC Codes_06.10.2026.csv"`.

Table 29: Codebook for Unique SIC Codes, Descriptions, and Column-Presence Flags (Post-Cleaning)

Field	Description
`sic_code`	The six-digit SIC code.
`sic_desc`	The description associated with the SIC code (following nomenclature cleaning).
`primary_sic_code...sic6_descriptions`	Boolean. TRUE if the code–description pair appears in the primary SIC code column; FALSE otherwise.
`sic_code...sic6_descriptions_sic`	Boolean. TRUE if the code–description pair appears in the first overflow SIC column; FALSE otherwise.
`sic_code_1...sic6_descriptions_sic1`	Boolean. TRUE if the code–description pair appears in the second overflow SIC column; FALSE otherwise.
`sic_code_2...sic6_descriptions_sic2`	Boolean. TRUE if the code–description pair appears in the third overflow SIC column; FALSE otherwise.
`sic_code_3...sic6_descriptions_sic3`	Boolean. TRUE if the code–description pair appears in the fourth overflow SIC column; FALSE otherwise.

Primary SIC Codes and Descriptions

For the implementation, refer to **PART B: Correct the SIC Nomenclature** in "Code/Clean Raw Data_Step 1_2026 Format.R". The data processed was the raw export `church_long_form_050926.csv`. Results were written to `"./Data/Results/KEEP LOCAL/From Process Data Update/Primary SIC Codes_06.10.2026.csv"`.

Table 30: Codebook for Unique Primary SIC Codes and Descriptions (Post-Cleaning)

Field	Description
`primary_sic_code`	The six-digit primary SIC code.
`sic6_descriptions`	The description associated with the SIC code (after nomenclature cleaning).

Step 1 Final Result

For the implementation, refer to "Code/Clean Raw Data_Step 1_2026 Format.R". The raw data from `"./Data/Raw/KEEP LOCAL/church_long_form_050926.csv"` were run through the nomenclature standardization and file type conversion outlined in **Step 1**. were run through the select nomenclature standardization and filetype conversion outlined in **Step 1**. Results were written to `"./Data/Results/KEEP LOCAL/From Clean Raw Data/Step 1_2026 Format/church_2026_form_standardized_06.10.2026.parquet"`.

Table 31: Codebook for new output fields produced by the evaluation.

Field	Description
`primary_naics6_code`	First six digits of the eight-digit NAICS code, representing the underlying NAICS classification (813110).
`naics6_descriptions`	Renamed form of the `naics8_descriptions` column.
`primary_naics2_code`	Last two digits of the eight-digit NAICS code. These are proprietary encodings added by Data Axle; no data dictionary was provided.

State-Level Shapefile for Basemap

For the implementation, refer to **SUBSECTION A2: Distribution of Characteristics Requiring Removal** in "Code/Clean Raw Data_Step 1_2026 Format.R". State-level census boundary files were incorporated using the `tigris` package [27]. Results were written to `"./Data/Results/From Clean Raw Data/Step 2_2026 Format/us_states_sf.shp"`.

Table 32: Codebook for `us_states_sf` 2023 Cartographic Boundaries Basemap

Field	Description
`STATEFP`	Two-digit state FIPS code (character).
`STATENS`	GNIS feature identifier for the state (character).
`GEOIDFQ`	Fully qualified geographic identifier (character).
`GEOID`	State geographic identifier (typically the state FIPS code as character).
`STUSPS`	USPS state abbreviation (e.g., "CA") (character).
`NAME`	State name (character).
`LSAD`	Legal/statistical area description code (character).
`ALAND`	Land area in square meters (numeric/integer).
`AWATER`	Water area in square meters (numeric/integer).
`geometry`	State boundary geometry as MULTIPOLYGON.

Missing Address Lines and PO Boxes By City

For the implementation, refer to **SUBSECTION A2: Distribution of Characteristics Requiring Removal** in "Code/Clean Raw Data_Step 2_2026 Format.R". The **Step 1** standardized data were summarized by city (as recorded on the form or, when needed, validated via ZIP code using the SimpleMaps Basic United States Cities Database (version 1.90) lookup table created with the custom function `build_zip_city_lookup()` [9]). These results were then joined to city geocoordinates from the `tigris` package [27]. Results showing missing `address_line_1` and the presence of PO Boxes were written separately to `"./Data/Results/From Clean Raw Data/Step 2_2026 Format/ABI with NA Addresses by City_08.07.2026.csv"` and `"~/ABI with PO Boxes by City_08.07.2026.csv"`.

Table 33: Codebook for Evaluation Summary Output Fields (Missing `address_line_1` and PO Box Datasets)

Field	Description
`state/city/zip_code`	Address elements associated with each entry.
`n_abi`	Number of unique ABIs represented in that city or zip code.
`[n|pct]_abi_any_na_address_line_1`	Count and percent of ABIs with at least one missing `address_line_1`.
`[n|pct]_abi_poBox`	Count and percent of ABIs with at least one PO Box entry.
`lon/lat`	TIGER/Line Shapefile geocoordinates associated with the city.

GEOID Prebuild Shapefiles

For the implementation, refer to **SUBSECTION A4: Build Precompiled TIGER/Line GeoPackages** in "Code/Clean Raw Data_Step 2_2026 Format.R". As described in the methods for LOOP PART D: Point-in-Polygon Spatial Assignment of Census Information, pulling in shapefiles using the `tigris` package was computationally prohibitive. Instead, relevant state-level shapefiles at the block and block-group level were manually downloaded from the Census Bureau TIGER/Line Shapefiles database (see Directory Specific Notes in the project GitHub repository for direct links to sources) [22]. These were then compiled into standardized shapefiles with layers for each decennial period (2000, 2010, and 2020) and written to the `"./Data/Results/Census Bureau TIGER Line Shapefiles"` directory.

Table 34: Codebook for State Block and Block-Group Shapefile Layers (2000/2010/2020)

Field	Description
`decennial_year`	The decennial census year associated with the boundary definition (2000, 2010, or 2020).
`geoid`	Full concatenated census geographic identifier, combining the state, county, tract, and block-group FIPS codes into a unique identifier for each boundary unit.
`statefp`	Two-digit state Federal Information Processing Series (FIPS) code identifying the state.
`countyfp`	Three-digit county FIPS code identifying the county within the state.
`tractce`	Six-digit census tract code identifying the census tract within the county.
`blkgrpce`	Single-digit block-group code identifying the block group within the census tract.
`geom`	Spatial geometry column containing the boundary polygons for each census unit, in the form of an `sf` geometry object.

Core Area CBSA/CSA Prebuild Shapefiles

For the implementation, refer to **SUBSECTION A4: Build Precompiled TIGER/Line GeoPackages** in "Code/Clean Raw Data_Step 2_2026 Format.R". As described in the methods for LOOP PART D: Point-in-Polygon Spatial Assignment of Census Information, pulling in shapefiles using the `tigris` package was computationally prohibitive. Instead,

relevant state-level shapefiles for the Metropolitan and Micropolitan Statistical Areas (CBSA) and Combined Statistical Areas (CSA) were manually downloaded from the Census Bureau TIGER/Line Shapefiles database (see Directory Specific Notes in the project GitHub repository for direct links to sources) [22]. These were then compiled into standardized shapefiles with layers for each decennial period (2000, 2010, and 2020) and written to the `"./Data/Results/Census Bureau TIGER Line Shapefiles"` directory.

Table 35: Codebook for Core Area Shapefiles (CBSA/CSA; 2007/2010/2020 Layers)

Field	Description
`vintage_year`	The decennial census year associated with the boundary definition (2007, 2010, or 2020).
`area_type`	Classification of the area as either a Core-Based Statistical Area (CBSA) or a Combined Statistical Area (CSA).
`area_code`	Official Census Bureau numeric code uniquely identifying the CBSA or CSA.
`area_level`	Hierarchical classification of the CBSA, indicating whether the area is Metropolitan or Micropolitan; CSA entries are `NA`.
`area_name`	Full official name of the statistical area as designated by the Census Bureau (e.g., Atlanta-Sandy Springs-Alpharetta, GA).
`area_states`	Concatenated string of state abbreviations for all states encompassed by the statistical area boundary.
`geom`	Spatial geometry column containing the boundary polygons for each census unit, in the form of an `sf` geometry object.

Core Area ZCTA Prebuild Shapefiles

For the implementation, refer to **SUBSECTION A4: Build Precompiled TIGER/Line GeoPackages** in "Code/Clean Raw Data_Step 2_2026 Format.R". As described in the methods for LOOP PART D: Point-in-Polygon Spatial Assignment of Census Information, pulling in shapefiles using the `tigris` package was computationally prohibitive. Instead, relevant state-level shapefiles for the ZIP Code Tabulation Areas (ZCTA) were manually downloaded from the Census Bureau TIGER/Line Shapefiles database (see Directory Specific Notes in the project GitHub repository for direct links to sources) [22]. These were then compiled into standardized shapefiles with layers for each decennial period (2000, 2010, and 2020) and written to the `"./Data/Results/Census Bureau TIGER Line Shapefiles"` directory.

Table 36: Codebook for ZCTA Shapefiles (2000/2010/2020 Layers)

Field	Description
`vintage_year`	The decennial census year associated with the boundary definition (2000, 2010, or 2020).
`area_type`	Classification of the area as ZCTA.
`area_code`	Official Census Bureau numeric code uniquely identifying the ZCTA.
`area_states`	Concatenated string of state abbreviations derived from a spatial join with the state-level TIGER/Line Shapefile for the respective decennial year.
`geom`	Spatial geometry column containing the boundary polygons for each census unit, in the form of an `sf` geometry object.

Step 2 Final Result

For the implementation, refer to "Code/Clean Raw Data_Step 2_2026 Format.R". The standardized records generated in **Step 1** were run through the sequential cleaning and validation steps outlined in **Step 2**. Results were compiled across a batch's array outputs and aggregated between batches into a single complete file in **PART C: Recompile Results from the HPC**, then written to `"./Data/Results/KEEP LOCAL/From Clean Raw Data/Step 2_2026 Format/Compiled by Batches/church_2026_form_validated_08.03.2026.db`.

Table 37: Codebook for new output fields produced by the evaluation.

Field	Description
address	Best-available address: verified_address (trimmed; excluding "No address match found"), else matched_address, else combined_address.
address_verified	USPS-verified address result when available; NA if no match is found or verification did not succeed.
address_matched	Result from agnostic address matching when performed (e.g., exact or fuzzy match). "Only one address" indicates matching was attempted but only one address was reported.
reported_address	Original address as reported, prior to cleaning or verification.
longitude/latitude	Best-available coordinates: verified coordinates when available, otherwise averaged raw coordinates (stored as numeric).
geolocation_verified	Indicator that geocoordinates were verified (definition depends on the verification step used).
geoid_[2000|2010|2020]	Full census GEOID assigned via point-in-polygon spatial join with block-group TIGER/Line boundaries for the corresponding decennial year.
geoid_match	Summary of GEOID assignment across years: "Matched" (all years), "Some matches not found" (partial), "Matches not found" (none), or "Not enough info" (insufficient coordinate data).
cbsa_code_[2007|2010|2020]	CBSA numeric code assigned via point-in-polygon spatial join for the corresponding vintage year.
cbsa_level_[2007|2010|2020]	CBSA level (Metropolitan vs. Micropolitan) for the corresponding vintage year.
csa_code_[2007|2010|2020]	CSA numeric code assigned via point-in-polygon spatial join for the corresponding vintage year.
zcta_[2000|2010|2020]	Five-digit ZCTA assigned via point-in-polygon spatial join with ZCTA TIGER/Line boundaries for the corresponding decennial year.

Step 2 Final Result - Import Quality Checks

For the implementation, refer to "Code/Clean Raw Data_Step 2_2026 Format.R". The standardized records generated in **Step 1** were run through the sequential cleaning and validation steps outlined in **Step 2**. Results were compiled across a batch's array outputs and aggregated between batches into a single complete file in **PART C: Recompile Results from the HPC**.

Selected outcomes and characteristics of these within-batch imports were generated to enable rapid confirmation of expected coverage and completeness. These were written with the main results to `"./Data/Results/KEEP LOCAL/From Clean Raw Data/Step 2_2026 Format/Compiled by Batches/church_2026_form_validated_08.03.2026.db`. Each batch's results are annotated with the first two digits of the batch number (`<XX>`).

Table 38: Codebook for `import_qc_<XX>$abi_check`

Field	Description
`array`	Batch array number from which the results were generated.
`file`	Source file name for the batch results.
`from/to`	ABI search-space index range defined under "SUBSECTION B1: Index Queue" in `Clean Raw Data_Step 2 HPC v2_2026 Format.R`.
`n_expected_unique`	Number of unique ABIs expected from the Step 1 data for the given index range.
`n_actual_unique`	Number of unique ABIs present in the batch results for the given index range.
`n_missing`	Number of ABIs absent from the batch results, computed as $\lvert$`setdiff(expected, actual)`$\rvert$.
`qc_pass`	Boolean: `TRUE` if $n\_missing = 0$, otherwise `FALSE`.

Table 39: Codebook for `import_qc_<XX>$[address_matched|address_verified|geoid_match|geolocation_verified]`

Field	Description
`array`	Batch array number from which the results were generated.
`value`	Unique outcome observed for the QC column (e.g., `"Exact match"`).
`n`	Number of records with the given outcome.
`n_addr`	Total number of addresses in the array.
`pct`	Percentage of addresses with the given outcome.

Table 40: Codebook for `import_qc_<XX>$na_census_boundaries`

Field	Description
`array`	Batch array number from which the results were generated.
`column`	Variable checked for `NA` values.
`n_addr`	Total number of addresses in the array.
`n_na`	Number of `NA` entries observed for the column.
`pct_na`	Percentage of addresses with an `NA` for that census boundary column.

Step 2 Algorithm Quality Check Datasets - Import Quality Checks

For the implementation, refer to "Code/Clean Raw Data_Step 2_2026 Format.R". The three quality control datasets — `address_qc`, `census_qc`, `geo_qc` — collected at the time of import were compiled from multiple HPC batches. A series of quality checks were run to confirm ABI coverage against the **Step 1** data. Each batch's results are annotated with the first two digits of the batch number (`<XX>`). These were written with the main results to the `"./Data/Results/KEEP LOCAL/From Clean Raw Data/Step 2_2026 Format/Compiled by Batches/` directory.

Table 41: Codebook for `import_qc_<XX>`

Field	Description
`array`	Batch array number from which the results were generated.
`file`	Source file name for the batch results.
`key`	Quality-check list (e.g., `qc1`, `qc2`) within the file from which the results were generated.
`from/to`	ABI search-space index range defined under "SUBSECTION B1: Index Queue" in `Clean Raw Data_Step 2 HPC v2_2026 Format.R`.
`n_expected_unique`	Number of unique ABIs expected from the Step 1 data for the given index range.
`n_actual_unique`	Number of unique ABIs present in the batch results for the given index range.
`n_missing`	Number of ABIs absent from the batch results, computed as $\lvert$`setdiff(expected, actual)`$\rvert$.
`qc_pass`	Boolean: `TRUE` if $n\_missing = 0$, otherwise `FALSE`.

Step 2 Algorithm Quality Check Datasets - Addresses

For the implementation, refer to "Code/Clean Raw Data_Step 2_2026 Format.R". The three quality control (QC) datasets — `address_qc`, `census_qc`, `geo_qc` — collected at the time of import were compiled from multiple HPC batches. The address QC dataset included three checks covering the address validation and matching process:

- **QC1:** A general indicator of whether API verification was allowed, which ABIs were verified, whether any addresses were matched, and two checks for duplications or other introductions and losses of results.
- **QC2:** Summarizes all quality check outcomes from address validation using the USPS API over unique ABIs and addresses.
- **QC3:** Summarizes all quality check outcomes from address matching over unique ABIs and addresses.

These were written with the main results to the `"./Data/Results/KEEP LOCAL/From Clean Raw Data/Step 2_2026 Format/Compiled by Batches/address_qc_08.05.2026.db`

directory.

Table 42: Codebook for `qc1`

Field	Description
`i`	Index of the `for` loop iteration (results are stored as lists).
`abi`	Business ID to which the QC results pertain.
`allow_usps_api`	Boolean: `TRUE` if USPS API usage was permitted during that array run via `verify_addresses`; otherwise `FALSE`.
`api_used`	Boolean: `TRUE` if the ABI entries are configured for USPS API usage via `do_api`; otherwise `FALSE`.
`verification_attempted`	Boolean: `TRUE` if `verified_address` is present. This reiterates `do_api` and should match it.
`match_attempted`	Boolean: `TRUE` if agnostic matching was attempted for any addresses for that ABI, indicated by the presence of `matched_address`.
`duplicates_induced`	Boolean: `TRUE` if any duplicates over `combined_address`, `anchor_year_min`, and `anchor_year_max` are detected; otherwise `FALSE`.
`any_addresses_line1_na`	Boolean: `TRUE` if any `address_line_1` for that ABI is `NA`; otherwise `FALSE`.
`new_not_in_old_addr_yr`	From `compare_tabs()`: indicates whether the cleaned data contains any unique `combined_address` and `archive_version_year` outcomes not present in the Step 1 standardized data.
`old_not_in_new_addr_yr`	Inverse of `new_not_in_old_addr_yr`: indicates whether the Step 1 standardized data contains any outcomes not present in the cleaned data.
`new_vs_old_differ`	Boolean: `TRUE` if the processed data differs in any way from the Step 1 standardized form; otherwise `FALSE`.

Table 43: Codebook for `qc2`

Field	Description
`i`	Index of the `for` loop iteration (results are stored as lists).
`abi`	Business ID to which the QC results pertain.
`address`	Best-available address: `verified_address` (trimmed; excluding "No address match found"), else `matched_address`, else `combined_address`.
`archive_versions_present`	String listing all years in which the address was observed.
`reported_address`	Original reported address, prior to any cleaning or verification.
`address_verified`	If the address was verified against the USPS API, this field is `TRUE`. If unverified but a match is found, the value indicates the string-matching mode (`"Override"` or `"Exact"`). If no `address_line_1` was present for matching, this is indicated with `"No address_line_1"`.
`ver_geolocation_test`	Geolocation test for matching attempts to verified addresses: `"Override"` if `"Exact"` matched; `TRUE`/`FALSE` if `"Fuzzy"` matched; `NA` if no match was attempted.
`usps_status`	Numeric status code summarizing the USPS API query interaction (e.g., `200`, `400`).
`usps_status_detail`	Status description corresponding to `usps_status` (e.g., `"200 Successful operation"`).
`attempt_succeeded`	Indicates whether verification succeeded on the first try, succeeded via fallback, or failed entirely.
`verified_address`	Verified address returned by the USPS API query.

Table 44: Codebook for `qc3`

Field	Description
`i`	Index of the `for` loop iteration (results are stored as lists).
`abi`	Business ID to which the QC results pertain.
`address`	Best-available address: `verified_address` (trimmed; excluding "No address match found"), else `matched_address`, else `combined_address`.
`archive_versions_present`	String listing all years in which the address was observed.
`reported_address`	Original reported address, prior to any cleaning or verification.
`address_matched`	If agnostic address matching was performed, this field records the outcome when matching succeeded via `"Exact matching"` or `"Fuzzy matching"`. `"Only one address"` indicates matching was attempted but only one address was reported.
`match_geolocation_test`	Geolocation test for agnostic matching attempts: `"Override"` if `"Exact"` matched; `PASS/FAIL` if `"Fuzzy"` matched. If `PASS` includes `"no Lon"`, `"no Lat"`, or `"no Lon/Lat"`, one record was missing that coordinate pair. `NA` if no match was attempted.
`matched_address`	Matched address result.

Step 2 Algorithm Quality Check Datasets - Census

For the implementation, refer to "Code/Clean Raw Data_Step 2_2026 Format.R". The three quality control (QC) datasets — `address_qc`, `census_qc`, `geo_qc` — collected at the time of import were compiled from multiple HPC batches. **QC1** and **QC2** apply the same tests over different columns, verifying whether the reported census boundaries correspond with any of the spatial join results, and whether the matched vintages align with the dates the address was filed.

These were written with the main results to the `"./Data/Results/KEEP LOCAL/From Clean Raw Data/Step 2_2026 Format/Compiled by Batches/census_qc_08.06.2026.db` directory.

Table 45: Codebook for `qc1`

Field	Description
`i`	Index of the `for` loop iteration (results are stored as lists).
`abi`	Business ID to which the QC results pertain.
`address`	Best-available address: `verified_address` (trimmed; excluding "No address match found"), else `matched_address`, else `combined_address`.
`archive_versions_present`	String listing all years in which the address was observed.
`census_[block|tract]/[county|fips]_code`	Original reported census boundaries recorded for the address (block/tract and county/FIPS codes, as available).
`n_address`	Number of times the `abi`–address combination was recorded.
`geoid_[2000|2010|2020]`	Full concatenated census GEOID assigned via point-in-polygon spatial join with block-group TIGER/Line boundaries for the corresponding decennial year.
`[fips|county|tract|block]_code_any_match`	Whether any original reported value matches any decennial value annotated via spatial joining: `"Matched"` if any do; `"Uncheckable"` if no original value was available; `"None"` if no matches.
`[fips|county|tract|block]_vintages`	Which decennial vintages matched: `"None"` if none matched, or `"Not reported"` if no original value was reported.
`[fips|county|tract|block]_vintages_aligned`	From `check_alignment()`: `TRUE` if matched vintages align with the years the address was reported; `FALSE` otherwise; `NA` if no comparison was available.
`block_type`	Level of the reported block boundary: `"None reported"` if missing; `"Blocks"` if four digits; `"Block groups"` if one digit; or `"Not the expected dimensions"` otherwise.

Table 46: Codebook for `qc2`

Field	Description
i	Index of the for loop iteration (results are stored as lists).
abi	Business ID to which the QC results pertain.
address	Best-available address: verified_address (trimmed; excluding "No address match found"), else matched_address, else combined_address.
archive_versions_present	String listing all years in which the address was observed.
n_address	Number of times the abi–address combination was recorded.
cbsa_[level|code]/csa_code	Original reported CBSA/CSA boundaries for the address (CBSA level and/or CBSA code, and CSA code, as available).
cbsa_code_[2007|2010|2020]	Official Census Bureau CBSA numeric code assigned via point-in-polygon spatial join for the corresponding vintage year.
cbsa_level_[2007|2010|2020]	CBSA classification (Metropolitan vs. Micropolitan) for the corresponding vintage year.
csa_code_[2007|2010|2020]	Official Census Bureau CSA numeric code assigned via point-in-polygon spatial join for the corresponding vintage year.
cbsa_[level|code] _code_any_match /csa_code_code_any_match	Whether any original reported value matches any vintage value annotated via spatial joining: "Matched" if any do; "Uncheckable" if no original value was available; "None" if no matches.
cbsa_[level|code]_vintages/ csa_code_vintages	Which vintages matched: "None" if none matched, or "Not reported" if no original value was reported.
cbsa_[level|code]_vintages _aligned/ csa_code_vintages_aligned	From check_alignment(): TRUE if matched vintages align with the years the address was reported; FALSE otherwise; NA if no comparison was available.

Step 2 Algorithm Quality Check Datasets - Geocoordinates

For the implementation, refer to "Code/Clean Raw Data_Step 2_2026 Format.R". The three quality control (QC) datasets — `address_qc`, `census_qc`, `geo_qc` — collected at the time of import were compiled from multiple HPC batches. Three quality checks were conducted over the geocoordinate verification process:

- **QC1**: Summarizes all quality check outcomes from address-based geocoding validation using the US Census Bureau over unique ABIs and addresses.
- **QC2**: Assesses the difference between the reported and validated geocoordinates.
- **QC3**: Examines the spread of geocoordinates among all records sharing the same ABI and address, and indicates whether this summary includes records where the address was verified or agnostically matched.

These were written with the main results to the `"./Data/Results/KEEP LOCAL/From Clean Raw Data/Step 2_2026 Format/Compiled by Batches/geo_qc_08.06.2026.db` directory.

Table 47: Codebook for `qc1`

Field	Description
`i`	Index of the `for` loop iteration (results are stored as lists).
`abi`	Business ID to which the QC results pertain.
`address`	Best-available address: `verified_address` (trimmed; excluding "No address match found"), else `matched_address`, else `combined_address`.
`archive_versions_present`	String listing all years in which the address was observed.
`n_address`	Number of times the `abi`–address combination was recorded.
`enough_geo`	Boolean: `TRUE` if both longitude and latitude were reported; otherwise `FALSE`.
`geolocation_verified`	Boolean: `TRUE` if an address-based geocoding match was retrieved from the US Census Bureau Geocoder API; otherwise `FALSE`.

Field	Description
`n_attempts`	Number of API query attempts made; each attempt uses a different benchmark and vintage in the query URL.
`query_statuses`	Summary of returned query statuses. `200` indicates a successful interaction even if no match was found. Attempts are separated by a vertical bar.
`all_200`	Boolean: `TRUE` if all benchmark/vintage pairs returned status `200` (interaction succeeded but no match found); otherwise `FALSE`.
`geo_matched_address`	Matched address returned by the geocoding database.
`matched_address_same`	Uses `find_similar_addresses(threshold = 0)` to check whether the addresses are exactly the same.
`matched_address_similar`	Uses `find_similar_addresses(threshold = 0.15)` to check whether the addresses are similar.
`benchmark/ vintage_input`	Specific reference database (benchmark and vintage) used to search for an address match during geocoding.
`address_verified`	If the address was verified against the USPS API, this field is `TRUE`. If unverified but a match is found, the value indicates the string-matching mode (`"Override"` or `"Exact"`). If no `address_line_1` was present for matching, this is indicated with `"No address_line_1"`.
`ver_geolocation_test`	Geolocation test for matching attempts to verified addresses: `"Override"` if `"Exact"` matched; `TRUE/FALSE` if `"Fuzzy"` matched; `NA` if no match was attempted.
`address_matched`	If agnostic address matching was performed, this field records the outcome when matching succeeded via `"Exact matching"` or `"Fuzzy matching"`. `"Only one address"` indicates matching was attempted but only one address was reported.
`match_geolocation_test`	Geolocation test for agnostic matching attempts: `"Override"` if `"Exact"` matched; `PASS/FAIL` if `"Fuzzy"` matched. If `PASS` includes `"no Lon"`, `"no Lat"`, or `"no Lon/Lat"`, one record was missing that coordinate pair. `NA` if no match was attempted.

Table 48: Codebook for `qc2`

Field	Description
`i`	Index of the `for` loop iteration (results are stored as lists).
`abi`	Business ID to which the QC results pertain.
`address`	Best-available address: `verified_address` (trimmed; excluding "No address match found"), else `matched_address`, else `combined_address`.
`archive_versions_present`	String listing all years in which the address was observed.
`[lat|lon]_abs_diff`	The absolute difference between the reported geocoordinate and the validated one.
`[lat|lon]_gt_002`	Boolean: `TRUE` if the absolute difference exceeds `0.002` degrees; otherwise `FALSE`. `NA` if there were no geocoordinates to compare.

Table 49: Codebook for `qc3`

Field	Description
`i`	Index of the `for` loop iteration (results are stored as lists).
`abi`	Business ID to which the QC results pertain.
`address`	Best-available address: `verified_address` (trimmed; excluding "No address match found"), else `matched_address`, else `combined_address`.
`n_address`	Number of times the `abi`–address combination was recorded.
`[lat|lon]_ [min|q1|median|mean|q3|max]`	Summary statistics describing the spread of geocoordinates reported for a unique address.
`[lat|lon]_spread_gt_02`	Boolean: `TRUE` if the spread of geocoordinates exceeds `0.02` degrees; otherwise `FALSE`. `NA` if there were no geocoordinates to compare.

Field	Description
any_address_verified	Boolean: TRUE if any address in these results came from a verified address; NA if no validation was used.
any_address_matched	Boolean: TRUE if any address in these results came from an agnostically matched address; NA if no matching was applied.

Step 3 Final Result

For the implementation, refer to **PART E: Reconstruct All Columns and Save Result** in "Code/Clean Raw Data_Step 3_2026 Format.R". The data processed were the **Step 2** results, run through all data cleaning steps outlined in **Step 3**. Results were written to "./Data/Results/KEEP LOCAL/From Clean Raw Data/Step 3_2026 Format/church_2026_form_wide_annotated_08.16.2026.parquet".

Table 50: Codebook for New Output Fields from Cleaning and Validation

Field	Description
2000:2025	Binary columns indicating years open and closed.
gap_filled	Boolean. TRUE if a qualifying gap between flanking 1's has been filled; FALSE otherwise.
n_gaps_filled	The number of gaps filled for that address.
avg_gap_len_filled	The average length, in years, of filled gaps; ranges from 1 to 3 sequential years.
n_moves	Number of times a given ABI/address was identified as a from_address, indicating a new move event.
to_address	The address to which the current entry relocated. These are the addresses quantified by the move variables.

Continued on next page

Field	Description
`transition_type`	Classifies the type of address transition. Gaps exceeding four consecutive years are classified as reopenings at a new location rather than moves. Addresses appearing multiple times temporally with a temporary new address may have multiple transitions associated per ABI/address.
`mean_dist_km`	The mean haversine distance (km) across all address transitions. For entries appearing more than once, this reflects the average distance across all moves; otherwise, it is the distance of the single move.
`max_dist_km`	The maximum haversine distance (km) across all address transitions. For entries appearing more than once, this reflects the largest single-move distance observed; otherwise, it is the distance of the single move.
`move_gt_[5mi|10mi|25mi]`	Boolean. TRUE if any of the moves exceeds 5, 10, or 25 miles, respectively.
`any_pobox`	Boolean. TRUE if a PO Box is present for either the `from_` or `to_address`; FALSE otherwise.
`primary_sic...primary_sic_desc, primary_sic... ...overflow_sic_desc_[1:11]`	Standardized primary and overflow SIC columns, made consistent across all ABI/address entries.
`buddhist_temple, christian_church, hindu_mandir, interfaith, jewish_synagogue, muslim_mosque, other_religion, sikh_gurdwara`	Boolean. TRUE if the address is classified under the given religion; FALSE otherwise.

Religious Classifiers Consistency Check

For the implementation, refer to **SUBSECTION B3: Assign Religious Classifiers** in "Code/Clean Raw Data_Step 3_2026 Format.R". Religious classifiers were compared against all primary and overflow SIC columns, with non-matching entries for the same ABI reported alongside. The data processed were the **Step 2** results. Comparison results were written to `"./Data/Results/From Clean Raw Data/Step 3_2026 Format/SIC Codes Consistency Check_XX_08.15.2026.xlsx"`, with the respective religion filled in place of XX.

Table 51: Codebook for SIC Description Consistency Sheets

Field	Description
`primary_pct100`	Sheet that filters the primary SIC code column for descriptions that are consistent across all addresses filed by an ABI.
`primary_pctlt100`	Sheet that filters the primary SIC code column for descriptions that are inconsistent across all addresses filed by an ABI. Each observed outcome is listed individually, rather than as pairs.
`overflow_pct100`	Sheet that is the same as `primary_pct100`, but applied to all deduplicated overflow column outcomes.
`overflow_pctlt100`	Sheet that is the same as `primary_pctlt100`, but applied to all deduplicated overflow column outcomes.
`x`	The SIC description (variable within each sheet).
`Freq`	The percentage of times the description appeared across all search hits (variable within each sheet).

Moves and Reopenings at New Addresses

For the implementation, refer to **PART D: Identifying and Quantifying Moves vs. Reopenings at New Location** in "Code/Clean Raw Data_Step 3_2026 Format.R". The data processed were the **Step 2** results. Results were written to `"./Data/Results/KEEP LOCAL/From Clean Raw Data/Step 3_2026 Format/Moves and Reopenings at New Address_08.16.2026.parquet"`, with the respective religion filled in place of XX.

Table 52: Codebook for Move-Event Quantification Data Frame

Field	Description
`abi`	The business ID to which the results relate.
`from_address`	The address from which the entry relocated. This is also listed as the primary address for that entry.
`to_address`	The address to which the entry relocated. These are the addresses quantified by the move variables.
`from_last_year`	The last year on file for the origin address.
`to_first_year`	The first year on file for the destination address.
`year_gap`	The number of years between the two filings.
`missing_years`	The number of years within the gap where no address was filed.
`dist_km`	The haversine distance between the two address coordinates, in km.
`move_gt_[5mi|10mi|25mi]`	Boolean. TRUE if the move distance exceeds 5, 10, or 25 miles, respectively.
`dist_flag`	Boolean. TRUE if the distance can be calculated (i.e., coordinates are non-missing and sufficient address information is available).
`transition_type`	Classifies the address change as `"Moved"` or `"Reopening (New Location)"` if the gap between filings exceeds 4 years.
`from_is_pobox`	Boolean. TRUE if the origin address is a PO Box; FALSE otherwise.
`to_is_pobox`	Boolean. TRUE if the destination address is a PO Box; FALSE otherwise.
`any_pobox`	Boolean. TRUE if either address is a PO Box; FALSE otherwise.

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